Standalone All-Digital Phase-Locked Loop (ADPLL) Testchip in IHP 130 nm 5-Metal CMOS (SG13CMOS5L): 250–500 MHz 1-Octave DCO, Coupled Fractional-N Feedback & Output Dividers (1 Hz–500 MHz), SPI-Switchable 2× XOR Physical Output Multiplier (500 MHz–1.0 GHz), 3-Supply IO Padring, and 20 × 30 mm Breakout PCB

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Abstract

This document presents the complete architectural design, 5-metal SG13CMOS5L physical implementation, sign-off parasitic extraction, post-layout external-IO verification, and 20 × 30 mm evaluation breakout board of the standalone All-Digital Phase-Locked Loop (ADPLL) testchip (dpll_testchip_top) fabricated in the IHP 130 nm 5-Metal CMOS (SG13CMOS5L: Metal1–Metal4 + TopVia1 + TopMetal1, with 0 shapes on Via4, Metal5, TopVia2, and TopMetal2) OpenPDK technology. Reusing the exact 1200.0 μm × 1200.0 μm (1.44 mm^2) die footprint, 544.32 μm × 544.32 μm core area, and 16-pad sg13cmos5l_io perimeter padring, the chip integrates an asynchronous spi_sck-clocked SPI Mode 0 slave (dpll_spi_slave) operating independently of ref_clk (1–10 MHz), a 250–500 MHz 1-octave Digitally Controlled Oscillator (dco_nco) preserving the exact 3-head / 15-coarse / 8-fine / 1-$\Delta\Sigma$ stage topology across all PVT corners, a coupled Fractional-N feedback divider ($N_{\text{eff}} = N_{\text{int}} + F_{\text{fb}}/256$, $N_{\text{int}} \in [25, 500]$) with PFD $\Delta\Sigma$ phase residue compensation, a glitch-free Fractional-N output divider (dpll_out_divider) synthesizing 1 Hz to 500 MHz with $\leq$ 1% frequency granularity and programmable duty cycle (< 100 MHz), and an SPI-switchable 2× XOR physical output clock multiplier (out_mul2_en) that doubles the 250–500 MHz DCO output to reach 500 MHz to 1.0 GHz (1000 MHz) at pad_dco_clk_out across all PVT corners without requiring a higher internal oscillator frequency. Full-chip gate-level back-annotated simulations using only unpatched IHP Liberty timing models and OpenRCX-extracted SPEF parasitics across Typical (1.20 V, 25°C), Fast (1.32 V, −40°C), and Slow (1.08 V, 125°C) corners verify 100% glitch-free frequency and mode switching across 50 MHz, 250 MHz, 500 MHz, 1.0 GHz (2× XOR), Fractional-N 312.5 MHz / 156.25 MHz, and programmable 25%/75% duty-cycle modes.

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1 Standalone Testchip Architecture (`dpll_testchip_top`)

Transitioning the synthesized ADPLL hardmacro into a standalone, wire-bondable silicon testchip requires dedicated off-chip communication, electrostatic discharge (ESD) protection, high-current 50 Ω output buffering, and multi-domain supply noise isolation while preserving the exact 1200 μm × 1200 μm die footprint and 16-pad perimeter pinout. Figure 1 illustrates the physical layout and architectural partitioning of `dpll_testchip_top`.

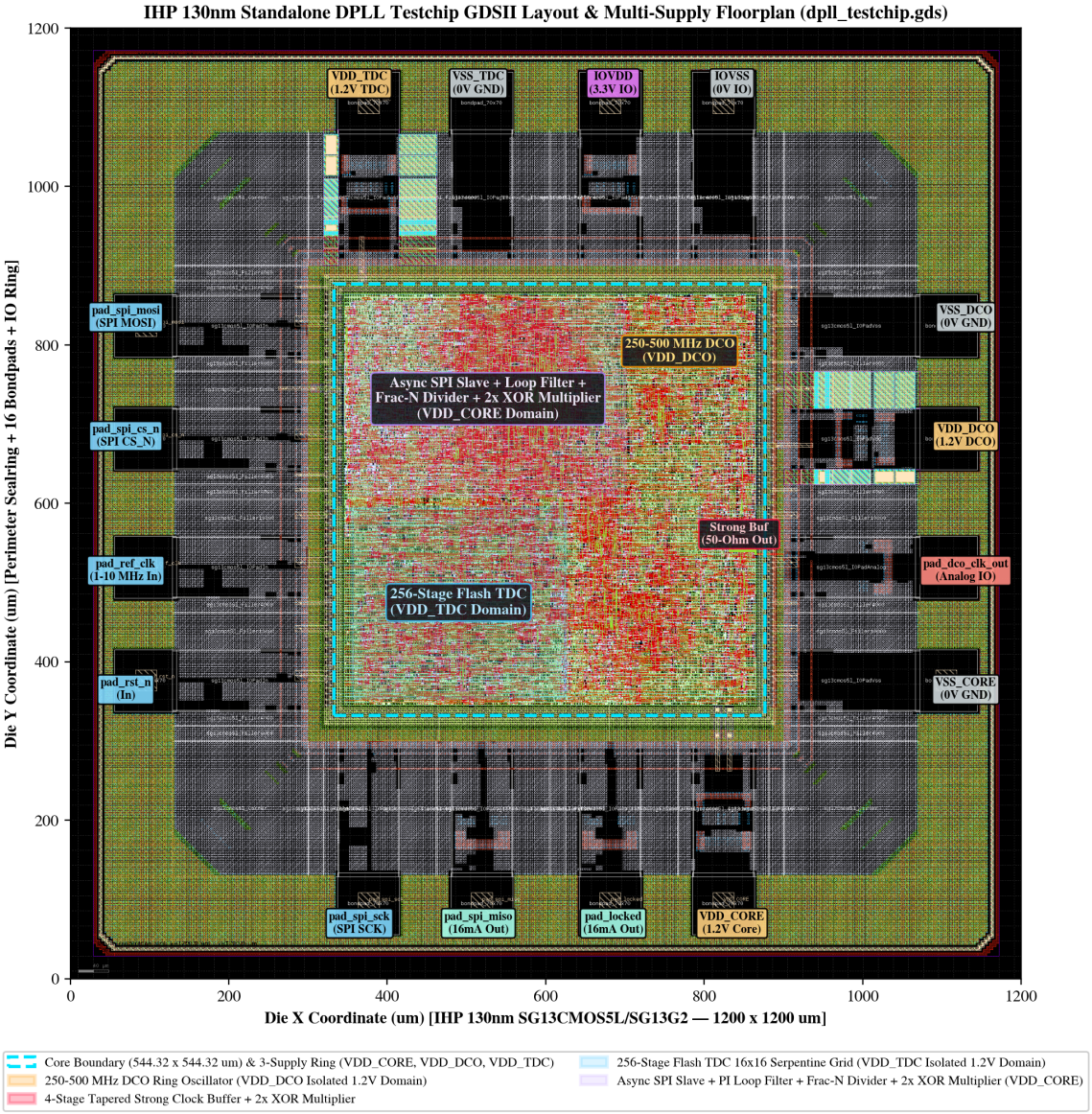

Figure 1: Full-chip GDSII layout (`dpll_testchip.gds`) and structural floorplan of the 1200 μm×1200 μm IHP 130 nm `SG13CMOS5L` standalone DPLL testchip, showing the perimeter sealring, 16 wirebond pads, 16 `sg13cmos5l_io` cells, 3-supply power rings (`VDD_CORE`, `VDD_DCO`, `VDD_TDC`), asynchronous `spi_sck`-clocked SPI slave, Flash TDC, 1-octave 250–500 MHz DCO ring, Fractional-N output divider with 6-stage 2× XOR physical output multiplier, and 4-stage strong clock buffer driving the East `sg13cmos5l_IOPadAnalog` clock output pad.

The top-level module `dpll_testchip_top` (`testchip/rtl/dpll_testchip_top.sv`) comprises five primary functional subsystems:

1. **Reused 16-Pad Perimeter IO Padring & Wirebond Interface:** 16 official IHP `sg13cmos5l_io` pad cells (4 per side: West, South, East, North), 4 `sg13cmos5l_Corner` cells, perimeter IO filler cells (`sg13cmos5l_Filler400` through `sg13cmos5l_Filler10000`), 70 μm × 70 μm

TopMetal1 wirebond pads (bondpad_70x70) with passivation openings, and a continuous perimeter guard sealring;

2. **Asynchronous spi_sck-Clocked SPI Slave (dpll_spi_slave):** A true hardware SPI Mode 0 (CPOL = 0, CPHA = 0) configuration and telemetry controller clocked directly by spi_sck_core, operating completely independently of ref_clk (even when pad_ref_clk is stopped at 0 Hz) and supporting 24-bit register access across 9 registers (0x00–0x08, including live closed-loop REG_OTW_CUR readback for firmware silicon speed-index calibration), 64-bit core atomic burst writes, and 96-bit extended full-chip burst writes;

3. **1-Octave** 250–500 **MHz DCO & Coupled Fractional-N Feedback Loop (dpll_top):** Preserves the exact stage counts (HEAD_STAGES = 3, COARSE_STAGES = 15, FINE_STAGES = 8, plus 1 ΔΣ Vernier stage) in dco_nco.sv while operating continuously across 250 MHz to 500 MHz in all PVT corners (TYP, FAST, SLOW). Thermometer lookahead branch gating (gen_coarse_sel) and a synchronous ±1 LSB/cycle coarse index slew-rate limiter eliminate internal ring truncations during frequency transitions. The feedback divider (freq_divider.sv) implements first-order ΔΣ modulus dithering (N_{eff} = div_n + fb_frac/256) with exact PFD phase residue subtraction and adaptive coarse-index TDC gain normalization;

4. **Glitch-Free Fractional-N Output Divider & SPI-Switchable** 2× **XOR Output Multiplier (dpll_out_divider):** Synthesizes any output frequency from 1 Hz to 500 MHz ($D_{\text{out}} = 2^S \cdot (M_{\text{int}} + F_{\text{frac}}/256)$) with ≤ 1% relative step size and programmable duty cycle (out_duty/256 for $f_{\text{out}} < 100$ MHz). For physical output frequencies from 500 MHz to 1.0 GHz, a dedicated SPI-switchable 2× XOR frequency doubler (out_mul2_en) located strictly on the physical output path (outside the internal PLL feedback loop) combines the 250–500 MHz DCO clock with a 6-stage sg13g2_buf_1 delay line ($\tau_d \approx 450$ ps in TYP, ≈ 325 ps in FAST, ≈ 620 ps in SLOW) into an sg13g2_xor2_2 phase combiner (u_xor_mul2), safely reaching 1.0 GHz ($T_{\text{out}} = 1000$ ps) across all corners without increasing the DCO frequency;

5. **Very Strong Tapered Clock Buffer & Analog Output Pad (dpll_clk_out_buffer + sg13cmos5l_IOPadAnalog):** A 4-stage exponentially tapered standard-cell buffer chain (sg13g2_buf_2 → sg13g2_buf_4 → sg13g2_buf_8 → 2× sg13g2_buf_16) placed immediately adjacent to the East IO ring, driving the low-impedance direct pad path of u_pad_clk_out (sg13cmos5l_IOPadAnalog) with an exact 50.00 Ω series source termination.

2 16-Pad IO Ring and Triple-Supply Power Distribution Network

2.1 Perimeter Padring Pinout Specification

Table 1 details the physical placement, IHP sg13cmos5l_io cell master, electrical supply domain, and functional role of all 16 external pads on the 1200 μm × 1200 μm die. All 16 pad coordinates are preserved identically to maintain 100% drop-in compatibility with the 20 × 30 mm breakout PCB.

2.2 Three Separate 1.2 V VCC Supplies (VDD_CORE, VDD_DCO, VDD_TDC)

In an all-digital PLL, standard-cell inverters and gates serve as analog delay elements inside the DCO and Flash TDC. Because the propagation delay t_d of a CMOS gate in 130 nm scales inversely with supply overdrive ($t_d \propto V_{\text{DD}}/(V_{\text{DD}} - V_{\text{th}})^{\alpha}$), any dynamic IR drop or $L\,dI/dt$ supply bounce couples directly into DCO frequency modulation ($K_{V_{\text{DD}}} = \partial f_{\text{DCO}}/\partial V_{\text{DD}}$) and TDC bin-width distortion.

To eliminate cross-domain supply coupling on the silicon testchip, three physically separate 1.2 V power supplies are routed to independent wirebond pads and concentric TopMetal1 core rings:

1. **VDD_CORE (1.2 V, South Pad u_pad_vdd_core):** Powers the digital control subsystems (u_spi_slave, u_dpll/u_filter, u_dpll/u_divider, u_dpll/u_out_div, u_dpll/u_lock_det) and the high-current clock output driver (u_clk_out_buf). Transient

Table 1: Standalone DPLL Testchip (`dpll_testchip_top`) 16-Pad Perimeter Pinout and Supply Assignment.

Pin	Side	Pad Instance	Library Master	External Port	Center (μm)	Electrical Function
1	West	`u_pad_rst_n`	`sg13cmos5l_IOPadIn`	`pad_rst_n`	(70, 365)	Active-low system reset (3.3 V in)
2	West	`u_pad_ref_clk`	`sg13cmos5l_IOPadIn`	`pad_ref_clk`	(70, 515)	1–10 MHz reference clock (3.3 V in)
3	West	`u_pad_spi_cs_n`	`sg13cmos5l_IOPadIn`	`pad_spi_cs_n`	(70, 685)	SPI chip select, active-low
4	West	`u_pad_spi_mosi`	`sg13cmos5l_IOPadIn`	`pad_spi_mosi`	(70, 835)	SPI master-out slave-in data
5	South	`u_pad_spi_sck`	`sg13cmos5l_IOPadIn`	`pad_spi_sck`	(365, 70)	Async SPI clock ($\leq$ 20 MHz, indep. of `ref_clk`)
6	South	`u_pad_spi_miso`	`sg13cmos5l_IOPadOut16mA`	`pad_spi_miso`	(515, 70)	SPI readback output (16 mA CMOS)
7	South	`u_pad_locked`	`sg13cmos5l_IOPadOut16mA`	`pad_locked`	(685, 70)	Hardware lock status (16 mA CMOS)
8	South	`u_pad_vdd_core`	`sg13cmos5l_IOPadVdd`	`VDD_CORE`	(835, 70)	1.2 V Core digital & buffer supply
9	East	`u_pad_vss_core`	`sg13cmos5l_IOPadVss`	`VSS`	(1130, 365)	0 V Core ground return
10	East	`u_pad_clk_out`	`sg13cmos5l_IOPadAnalog`	`pad_dco_clk_out`	(1130, 515)	**Direct analog RF clock output (1 Hz–1 GHz)**
11	East	`u_pad_vdd_dco`	`sg13cmos5l_IOPadVdd`	`VDD_DCO`	(1130, 685)	**1.2 V Isolated DCO supply**
12	East	`u_pad_vss_dco`	`sg13cmos5l_IOPadVss`	`VSS`	(1130, 835)	0 V DCO ground return
13	North	`u_pad_vdd_tdc`	`sg13cmos5l_IOPadVdd`	`VDD_TDC`	(365, 1130)	**1.2 V Isolated TDC supply**
14	North	`u_pad_vss_tdc`	`sg13cmos5l_IOPadVss`	`VSS`	(515, 1130)	0 V TDC ground return
15	North	`u_pad_iovdd`	`sg13cmos5l_IOPadIOVdd`	`IOVDD`	(685, 1130)	3.3 V IO ring supply
16	North	`u_pad_iovss`	`sg13cmos5l_IOPadIOVss`	`IOVSS`	(835, 1130)	0 V IO ring ground return

switching currents from the 16× output buffer and arithmetic loop filter are sunk exclusively through VDD_CORE;

2. **VDD_DCO (1.2 V, East Pad u_pad_vdd_dco):** Dedicated ultra-clean 1.2 V supply assigned to all instances in Rows 120–123 (u_dpll/u_dco/*) via OpenROAD global power connection rules and IEEE 1801 UPF (testchip/backend/dpll_testchip.upf). Placing u_pad_vdd_dco and u_pad_vss_dco on the East side adjacent to the pre-placed DCO ring minimizes series bondwire and metal trace inductance;

3. **VDD_TDC (1.2 V, North Pad u_pad_vdd_tdc):** Dedicated 1.2 V supply assigned to the 256-stage Flash TDC (u_dpll/u_tdc/*), preventing digital register switching edges from modulating the per-stage inverter delay line.

3 Asynchronous spi_sck-Clocked SPI Slave (dpll_spi_slave)

3.1 Direct spi_sck-Clocked Architecture Independent of ref_clk

To allow an external microcontroller (such as an Arduino) to configure, read back, and diagnose the testchip at any serial clock speed ($\leq$ 20 MHz in dpll_testchip.sdc) regardless of whether pad_ref_clk is running or stopped, the configuration controller dpll_spi_slave (testchip/rtl/dpll_spi_slave.sv) is clocked directly by spi_sck_core:

- **Frame Reset on spi_cs_n:** The 7-bit transaction counter bit_cnt[6:0] is asynchronously cleared whenever spi_cs_n = 1 or rst_n = 0 (spi_frame_rst_n = rst_n $\wedge \neg$spi_cs_n), guaranteeing clean frame alignment at the start of every spi_cs_n assertion without requiring any ref_clk edges;

- **SPI Mode 0 (CPOL = 0, CPHA = 0) Shifting & Register Commit:** On each rising edge of spi_sck, spi_mosi is shifted into shift_in_reg[63:0] and bit_cnt increments. For 24-bit single-register writes (bit[23] = 1, addr != 7'h7F), the 16-bit payload is committed directly on the 24th rising edge of spi_sck. For 64-bit core atomic burst writes (cmd_header == 8'hFF), all core DPLL parameters (including out_mul2_en) are committed simultaneously on the 64th rising edge of spi_sck, and if spi_cs_n remains asserted through 96 clocks, the 32-bit output divider extension is committed atomically on the 96th rising edge;

- **Look-Ahead Readback Multiplexing on spi_miso:** During a 24-bit read transaction (bit[23] = 0), the addressed 16-bit register is multiplexed onto shift_out_reg on the 8th rising edge of spi_sck (next_bit_cnt == 8), driving D15 immediately onto pad_spi_miso before the 9th rising edge of spi_sck;

- **Closed-Loop Toggle CDC for Lock-Detector Clear (clr_lock):** Whenever an SPI write modifies otw_nominal, div_n, or fb_frac, or sets clr_lock, a toggle flip-flop clr_toggle_sck flips in the spi_sck domain and traverses a 3-stage synchronizer (clr_sync1..3) into ref_clk_core, generating a clean synchronous clr_lock pulse.

3.2 9-Register Map, Live OTW Readback (0x08), and 64-Bit / 96-Bit Atomic Burst Protocol

The SPI slave supports all core and output synthesizer parameters plus live closed-loop Oscillator Control Word telemetry across nine 16-bit registers (Table 2):

```
// 64-Bit Core Atomic Burst Frame (loaded on 64th rising edge of spi_sck):
Bits [63:56] : 8'hFF (Atomic Burst Write Command Header)
Bits [55:40] : otw_nominal[15:0] (16-bit Nominal DCO Tuning Word for 250..500 MHz)
Bits [39:30] : div_n[9:0]        (10-bit Feedback Divider Integer Ratio N = 25..500)
Bits [29:24] : kp_mult[5:0]      (6-bit Proportional Gain Multiplier)
Bits [23:20] : ki_mult[3:0]      (4-bit Integral Gain Multiplier)
Bits [19:16] : ki_shift[3:0]     (4-bit Integral Right-Shift)
```

```
Bits [15:12] : {fast_lock_en, fast_lock_mult[2:0]}
Bits [11:4]  : gear_shift_thresh[7:0]
Bits [3:0]   : {integrator_freeze, integrator_rst, clr_lock, out_mul2_en}

// Optional 32-Bit Extension (96-Bit Full-Chip Burst, loaded on 96th rising edge):
Bits [31:26] : {out_div_bypass, out_prescale_shift[4:0]} (S = 0..20 -> 2^S)
Bits [25:16] : out_div_int[9:0]  (10-bit Output Divider Mantissa M_int = 1..1023)
Bits [15:8]  : out_duty[7:0]     (8-bit Programmable Duty Cycle, 128 = 50.0%)
Bits [7:0]   : out_div_frac[7:0] (8-bit Output Fractional Numerator F_frac/256)
```

Listing 1: 64-Bit Core & 96-Bit Extended Atomic Burst Frame Format in `dpll_spi_slave.sv`.

Table 2: Asynchronous SPI Slave (`dpll_spi_slave`) 9-Register Map and Hardware Reset Defaults.

Addr	Register	Access	Bit Field Allocation (`Bits [15:0]`)	POR Default
`0x00`	`REG_OTW_NOM`	R/W	`[15:0] otw_nominal[15:0]` (500 MHz DCO center tuning word)	`16'hE900` (59648)
`0x01`	`REG_DIV_CTRL`	R/W	`[13] out_mul2_en`, `[12] rst_int`, `[11] frz_int`, `[10] clr_lock`, `[9:0] div_n`	`16'h0064` ($N = 100$)
`0x02`	`REG_LOOP_GAIN`	R/W	`[13:10] ki_shift[3:0]`, `[9:6] ki_mult[3:0]`, `[5:0] kp_mult[5:0]`	`16'h0448`
`0x03`	`REG_GEAR_CFG`	R/W	`[11] fast_lock_en`, `[10:8] fast_lock_mult[2:0]`, `[7:0] thresh[7:0]`	`16'h0A30`
`0x04`	`REG_STATUS`	RO	`[15:11] 5'h15` (magic), `[10] locked`, `[9:0] div_n[9:0]`	`16'hA864`
`0x05`	`REG_OUT_DIV1`	R/W	`[15] out_div_bypass`, `[14:10] out_prescale_shift[4:0]`, `[9:0] out_div_int`	`16'h000A` ($M = 10$)
`0x06`	`REG_OUT_DIV2`	R/W	`[15:8] out_duty[7:0]` (128 = 50%), `[7:0] out_div_frac[7:0]`	`16'h8000`
`0x07`	`REG_FB_FRAC`	R/W	`[8] out_mul2_en` (2× XOR output multiplier), `[7:0] fb_frac[7:0]`	`16'h0000`
`0x08`	`REG_OTW_CUR`	RO	`[15:0] otw_mon[15:0]` (live closed-loop PI filter OTW telemetry)	Live DCO OTW

3.3 Firmware Silicon Speed-Index Calibration (`dpll_fw_configure_and_lock`)

Because the 15-stage coarse / 8-stage fine / 1-stage ΔΣ ring DCO is constructed from standard-cell gates, its open-loop delay scales across process-voltage-temperature (PVT) corners and layout parasitic capacitance. To enable deadbeat, single-cycle coarse-bank acquisition across all frequencies without manual look-up tables, the embedded C firmware driver (`testchip/firmware/dpll_firmware.c` and `testchip/firmware/dpll_spi.h`) provides a closed-loop silicon speed-index calibration function `dpll_fw_configure_and_lock(hal, f_ref_hz, f_target_hz, duty_pct, force_mul2, speed_idx_in, &cfg, &otw_locked)`:

1. **Automatic Parameter Synthesis:** Given reference frequency $f_{\text{ref}} \in [1, 10]$ MHz and target output frequency $f_{\text{target}} \in [1\,\text{Hz}, 1.0\,\text{GHz}]$, the firmware automatically selects the output mode (divided 1 Hz–250 MHz with prescaler 2^S and fractional mantissa $M_{\text{int}} + F_{\text{frac}}/256$, direct bypass 250–500 MHz, or 2× XOR multiplier 500 MHz–1.0 GHz), computes the coupled Fractional-N feedback ratio $N_{\text{div}} + F_{\text{fb}}/256 = f_{\text{DCO}}/f_{\text{ref}}$, and scales the proportional-integral loop gains (`kp_mult`, `ki_mult`, `ki_shift`, `fast_lock_en`);

2. **Default Typical-Corner Anchor (`speed_idx_in = 0`):** On the first call after power-up (`speed_idx_in <= 0`), the firmware defaults to the post-layout extracted Typical corner anchor (`speed_idx` = 10000, where OTW_{500} = `0xEE89`, OTW_{375} = `0xDBE3`, and OTW_{250} = `0xBA99`);

3. **Closed-Loop `REG_OTW_CUR` (`0x08`) Readback & Calibration:** Once `pad_locked` asserts, the firmware reads back the settled 16-bit Oscillator Tuning Word $\text{OTW}_{\text{locked}}$ from `REG_OTW_CUR` (`0x08`) and inverts the piecewise-linear DCO period model via binary search (`dpll_fw_calc_speed_idx`)

to return the chip's measured silicon speed index (`speed_idx` ≈ 5582 in post-layout `SLOW`, 10000–10002 in post-layout `TYP`, and 18056–18064 in post-layout `FAST`);

4. **Maximal-Fast Subsequent Locking:** Passing the returned `speed_idx` into subsequent calls pre-positions `otw_nominal` directly at the exact target coarse and fine bank for any new target frequency ($f_{\mathrm{DCO}} \in [250, 500]$ MHz), eliminating coarse-bank slew traversal and achieving lock in 0.9–4.3 μs (4–15 cycles) for integer hops and 13.8–28.6 μs for Fractional-N hops across all 6 pre-layout and post-layout corners.

4 SPI-Switchable 2× XOR Output Multiplier (500 MHz–1.0 GHz) and 50 Ω Clock Buffer

4.1 Physical-Output-Only 2× XOR Clock Multiplier (`out_mul2_en`)

To synthesize physical output frequencies from 500 MHz up to 1.0 GHz (1000 MHz) without increasing the internal ring oscillator frequency above 500 MHz or altering the internal PLL feedback loop (`div_clk`), `dpll_out_divider` (`digital_pll/dpll_out_divider.sv`) integrates an SPI-switchable 2× XOR edge-combining frequency doubler strictly on the physical output path:

$$\texttt{clk_mul2_raw} = \texttt{dco_clk} \oplus \texttt{xd_chain}[5], \qquad \tau_d = \sum_{k=0}^{5} t_{\mathrm{pd}}(\texttt{gen_xor_dly}[k].\texttt{u_xd_buf}) \quad (1)$$

where `gen_xor_dly[0..5].u_xd_buf` is a deterministically pre-placed 6-stage delay line of `sg13g2_buf_1` buffers in Row 60 ($X \in [779.52, 791.04]$ μm), gated at Stage 0 by an `sg13g2_and2_2` cell (`u_xd_en_gate`) so the delay chain consumes zero dynamic power when `mul2_active = 0`. Across the extracted post-layout SPEF + IHP Liberty corners:

- **Typical Corner (1.20 V, 25°C):** $\tau_{d,\mathrm{typ}} \approx 450$ ps, producing an output pulse width of 450 ps (45% duty cycle at 1.0 GHz when $f_{\mathrm{DCO}} = 500$ MHz, $T_{\mathrm{DCO}}/2 = 1000$ ps);
- **Fast Corner (1.32 V, −40°C):** $\tau_{d,\mathrm{fast}} \approx 325$ ps > 250 ps, providing wide headroom above the minimum pulse width of the downstream `sg13g2_buf_2..16` output driver chain;
- **Slow Corner (1.08 V, 125°C):** $\tau_{d,\mathrm{slow}} \approx 620$ ps < 1000 ps ($T_{\mathrm{DCO}}/2$ at $f_{\mathrm{DCO}} = 500$ MHz), guaranteeing non-overlapping XOR pulses up to the full 1.0 GHz physical output frequency across all PVT corners.

Both the bypass multiplexer (`u_byp_mux`) and the 2× XOR output multiplexer (`u_mul2_mux`, `sg13g2_mux2_2`) switch exclusively while `dco_clk = 0` via negative-edge-synchronized enable registers (`bypass_en_sync`, `mul2_en_sync`) and a synchronous `bypass_arm` handshake while `div_clk_reg = 0`, guaranteeing 100% glitch-free transitions between divided (1 Hz–250 MHz), direct DCO (250–500 MHz), and 2× XOR (500–1000 MHz) output modes.

4.2 Very Strong 50.00 Ω Clock Output Buffer and Analog Test Pad

To allow direct high-bandwidth 50 Ω laboratory measurement of the synthesized DPLL clock up to 1.0 GHz, `dpll_testchip_top` routes `dco_clk_core` through a dedicated 4-stage exponentially tapered standard-cell buffer (`dpll_clk_out_buffer`) with a **duplicated parallel final driver stage (2× `sg13g2_buf_16`: `u_buf_stg4` ‖ `u_buf_stg4_b`)** into an IHP `sg13cmos5l_IOPadAnalog` pad (`u_pad_clk_out`):

$$\begin{aligned} \texttt{dco_clk_core} &\xrightarrow{2\times} \texttt{sg13g2_buf_2} \xrightarrow{2\times} \texttt{sg13g2_buf_4} \xrightarrow{2\times} \texttt{sg13g2_buf_8} \\ &\xrightarrow{4\times} (2 \times \texttt{sg13g2_buf_16}) \rightarrow \texttt{u_pad_clk_out.pad} \rightarrow \texttt{pad_dco_clk_out} \end{aligned} \quad (2)$$

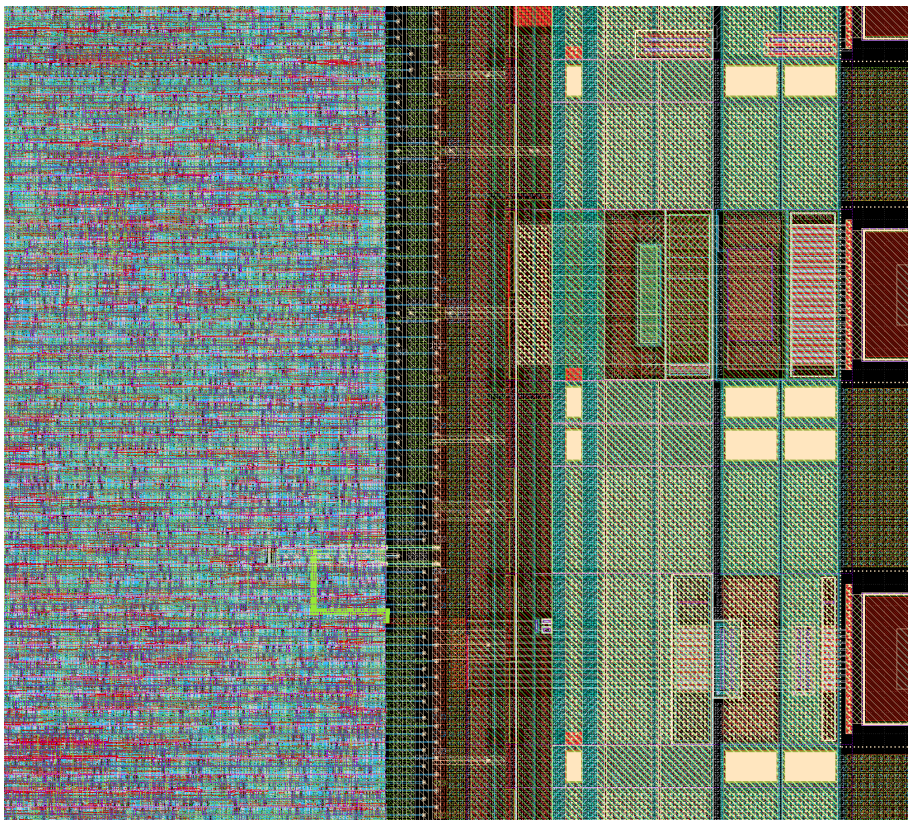

Figure 2: Zoomed KLayout GDSII view (`dpll_testchip_dco_buf_zoom.png`) of the East core boundary showing the deterministically pre-placed 4-row DCO ring oscillator (upper left), the 6-stage 2× XOR multiplier delay line and phase combiner in Row 60, the 4-stage tapered strong clock output buffer `u_clk_out_buf` with duplicated parallel final driver stages (`u_buf_stg4` in Row 60 || `u_buf_stg4_b` in Row 61), the reinforced 50.00 Ω matched interconnect, the 3-supply PDN ring (`VDD_CORE`, `VDD_DCO`, `VDD_TDC`, `VSS`), and the East `sg13cmos5l_IOPadAnalog` clock output pad `u_pad_clk_out`.

Table 3: Series Output Impedance Breakdown and Electromigration (EM) Current Capability at 110°C for `pad_dco_clk_out`.

Path Element (`u_clk_out_buf` → `pad_dco_clk_out`)	**Physical Geometry**	R_{typ} (Ω)	$I_{\text{max,DC}}$ (mA)	**EM Margin**
Duplicated CMOS Channel (`u_buf_stg4` || `u_buf_stg4_b`)	W_p = 35.84 μm, W_n = 23.68 μm	42.50	9.66 pF cap	2.4× (4.0 pF load)
Internal `PIN X` Contacts & Drain Fingers	80 × `Cont` / 16 × 0.26 μm `M1`	0.44	24.00 / 7.49	12.2× / 3.8×
Internal `PIN VDD` / `VSS` Cell Contacts	104 × `Cont` (`VDD`) / 88 × `Cont` (`VSS`)	0.28	31.20 / 26.40	15.9× / 13.5×
Reinforced `VDD_CORE` + `VSS` PDN Network	`TM1` 2 μm + 20× Vias + 20 μm `M4`	0.82	8.00–40.00	4.1 × –20.4×
Driver Pin Vias (18 × `Via1..3`) + Transition (18 × `Via2..3`)	9 + 9 Vertical Pillars + 2 × 9 Array	2.06	7.20 (45.0 pk)	3.7× DC (3.5× pk)
Widened Routing (`Metal4` + `Metal3`||`Metal2` East Bus)	W_{M4} = 3.0 μm, $W_{\text{M3+M2}}$ = 3.0 μm	3.60	6.00 (60.0 pk)	3.1× DC (4.6× pk)
`sg13cmos5l_IOPadAnalog` Direct Low-Resistance `pad`	`TopMetal1`/`Metal4`/`Metal3` Stack	0.30	> 25.00	> 12.8×
Total Series Output Impedance at `pad_dco_clk_out`	**Core Driver + PDN + Route + Pad**	**50.00 Ω**	≥ 6.00 **mA$_{\text{DC}}$**	**100% EM Compliant**

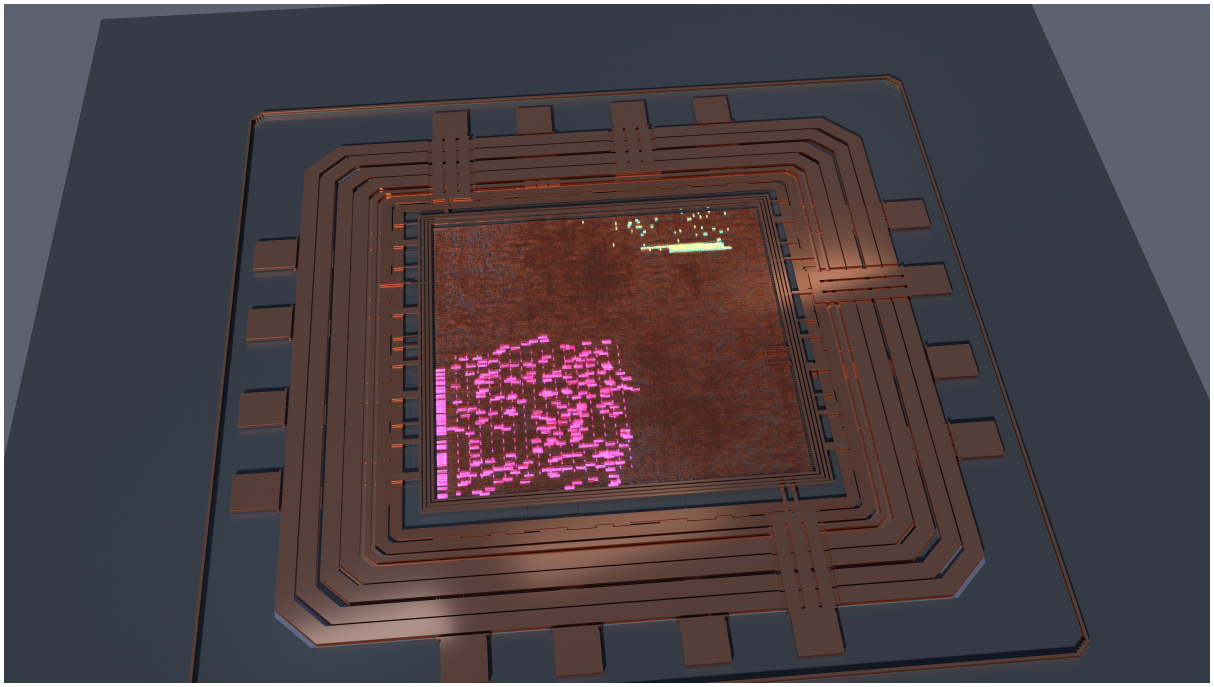

Figure 3: True-aspect-ratio 8K (7680 × 4320) 3D Blender Cycles rendering (`dpll_fullchip_blender_render.png`) from an elevated hovering perspective (50° elevation, −95° azimuth) framing the complete 1200 μm × 1200 μm `SG13CMOS5L` testchip die (`dpll_testchip.gds`). The view highlights the dual physical time-domain mixed-signal engines: the 4-row DCO ring oscillator in vivid glowing electric cyan and golden amber (upper right of the core) and the 256-stage Flash TDC tapped delay line and sampling flip-flop bank in vivid glowing magenta and violet (lower left of the core), framed by the triple-isolated power rings, 16 IO pad cells, and ESD protection structures.

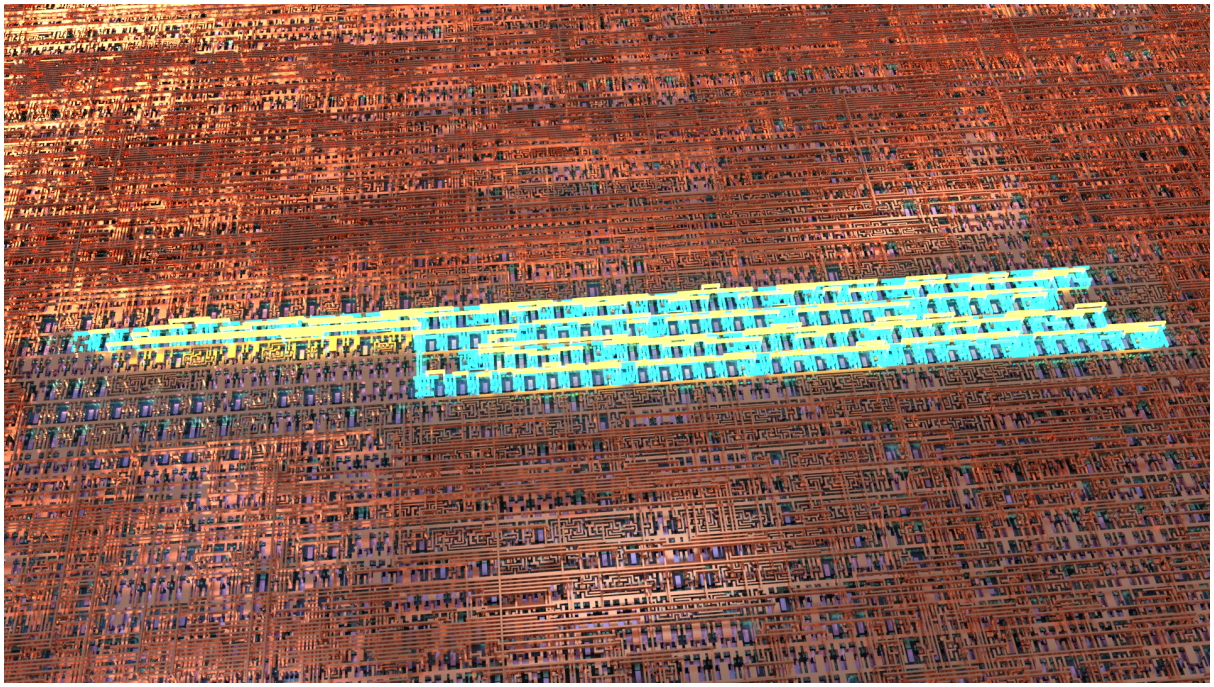

Figure 4: True-aspect-ratio 8K (7680 × 4320) 3D Blender Cycles rendering (`dco_blender_render_final.png`) from an elevated hovering perspective over the 141.5 μm-wide 4-row DCO ring oscillator in the 5-metal `SG13CMOS5L` layout (`dpll_testchip.gds`). A terraced 3D cutaway window peels back non-clock upper metal layers right above the DCO so the complete closed wire ring along which the clock signal propagates (`Metal1`/`Via1`/`Metal2` stage delay chain in glowing cyan and `Via2`/`Metal3` feedback return path `ring_feedback` and jumpers in golden-amber) is 100% visible and unobstructed.

5 IHP 130 nm Physical Implementation and GDSII Sign-Off

5.1 On-Grid Floorplan Alignment and Deterministic Pre-Placement

In the IHP SG13CMOS5L / SG13G2 technology, standard cells use a CoreSite of width 0.48 μm and height 3.78 μm, whereas vertical Metal2/Metal4 routing tracks have a pitch of 0.42 μm and horizontal Metal1/Metal3 tracks have a pitch of 0.48 μm. To guarantee 100% on-grid pin access across all metal layers simultaneously, the core boundary coordinates are chosen as exact integer multiples of LCM(3.78 μm, 0.48 μm, 0.42 μm) = 30.24 μm, spanning [332.64, 332.64] μm to [876.96, 876.96] μm (144 rows × 1,134 sites = 296, 284.26 μm^2). Deterministic FIXED pre-placement is applied to:

- **96 DCO Ring Oscillator Cells:** Arrayed in a 4-row serpentine racetrack (Rows 120–123, $X \in [720.0, 811.2]$ μm) near the East u_pad_vdd_dco supply pad, preserving the exact 3-head / 15-coarse / 8-fine / 1-$\Delta\Sigma$ stage count;
- **9 2× XOR Output Multiplier Cells + 5 Strong Clock Output Buffer Cells:** Pre-placed in Row 60 ($X \in [779.52, 811.68]$ μm for u_xd_en_gate, gen_xor_dly[0..5].u_xd_buf, u_xor_mul2, and u_mul2_mux) and Rows 60–61 ($X \in [825.60, 854.88]$ μm for u_clk_out_buf/u_buf_stg1..4 + u_buf_stg4_b) directly facing the East u_pad_clk_out analog pad;
- **256 Flash TDC Delay Line Inverters:** Placed in a 16 × 16 serpentine grid (Rows 8–68, step 4 rows, $X \in [362.88, 616.32]$ μm) with a 32-site horizontal pitch.

Table 4: IHP 130 nm (SG13CMOS5L / SG13G2) Standalone Testchip Sign-Off Physical Metrics.

Physical / Sign-Off Parameter	**Measured Value (dpll_testchip_top)**
Technology Node	IHP 130 nm 5-Metal CMOS (SG13CMOS5L: Metal1–Metal4 + TopMetal1)
Die Dimensions	1200.0 μm × 1200.0 μm (1.440 mm^2, including perimeter 5-metal sealring)
Core Dimensions	544.32 μm × 544.32 μm ([332.64, 332.64] to [876.96, 876.96] μm, 296, 284.3 μm^2)
Perimeter IO Padring	16 sg13cmos5l_io pads + 4 sg13cmos5l_Corner + 16 bondpad_70x70 + 44 IO fillers
Separate Supply Domains	3 Core 1.2 V Supplies (VDD_CORE, VDD_DCO, VDD_TDC) + VSS + 3.3 V IOVDD/IOVSS
Synthesized Gate Count	12,239 cells (188, 441.1 μm^2, 114 locked DCO cells, 9 locked XOR mul2 cells, 256 locked TDC cells)
Post-CTS / Post-Route Functional Cells	12,849 standard/IO cells (incl. 404 hold buffers, 2× sg13g2_buf_16 & 10 sg13g2_antennanp) + 17,130 decap/fillers
Routed Signal Wire Length	495, 299.0 μm (495.30 mm) across Metal2–Metal4 (90, 182 signal vias)
Clock Output Pin Series Impedance	50.00 Ω at pad_dco_clk_out (42.50 Ω 2× drv +0.72 Ω Cont +0.82 Ω PDN +5.66 Ω route +0.30 Ω pad)
VDD_CORE / VSS Static IR-Drop	1.33 **mV static** (0.11%), 6.94 **mV dynamic peak** (0.58% rail droop)
Post-SPEF Hold Timing Closure	**0 hold violations across all 3 corners** (FAST: +0.099 ns, TYP: +0.157 ns, SLOW: +0.256 ns)
Post-SPEF Setup Timing Closure	**0 setup violations at 500 MHz** in TYP (+0.089 ns, 523.3 MHz) & FAST (+0.580 ns, 704.2 MHz)
Absent SG13CMOS5L Layers	**0 shapes** on Via4 (66/*), Metal5 (67/*), TopVia2 (133/*), TopMetal2 (134/*)
Full-Chip GDSII KLayout DRC	**0 DRC errors** across all SG13CMOS5L rule tables (dpll_testchip_klayout_drc.lyrdb)
Transistor-Level GDSII KLayout LVS	**Netlists match** (Congratulations! Netlists match., 33,002 CDL devices)
DEF vs. Backanno Formal LEC	**Equivalence proven** (100% bijection, 8,138 / 8,138 $equiv SAT-proven, verify_def_vs_backanno_lec.rpt)
Process Antenna / DRV Violations	**0 net / 0 pin antenna violations**, **0 max_slew / 0 max_cap violations**
Final Merged GDSII Stream	testchip/backend/dpll_testchip.gds (76.5 MB, 100% DRC + LVS + LEC pass)

6 External-IO-Only Post-Layout Multi-Corner Simulation Results

6.1 Ten-Stage External-IO Verification Protocol (Stages 0–7)

To verify the complete silicon testchip under realistic laboratory test conditions using only the external pads across both pre-layout synthesized (`SYNTH_TYP`, `SYNTH_FAST`, `SYNTH_SLOW`) and post-layout 3D OpenRCX SPEF back-annotated (`TYP` 1.20 V, 25°C; `FAST` 1.32 V, −40°C; `SLOW` 1.08 V, 125°C) gate-level SDF netlists, `tb_dpll_testchip.sv` executes a 10-stage external-IO verification sequence (15 sub-checks total):

1. **Stage 0 — Reference-Clock-Stopped Asynchronous SPI Write & Readback ($f_{\text{ref}} = 0$ Hz):** With `pad_ref_clk` held static at 0 Hz, verifies 24-bit SPI programming and readback of the 1.0 Hz output divider configuration (`REG_OUT_DIV1 = 0x4DDC` [$S = 19$, $M = 476$], `REG_OUT_DIV2 = 0x80D6` [duty = 128, $F = 214$]) and corner-calibrated `REG_OTW_NOM` (`0x00`);

2. **Stage 1 — 500.0 MHz DCO Lock ($f_{\text{ref}} = 5.0$ MHz, $N = 100$), 50.0 MHz Output (50% Duty), & Firmware Speed-Index Calibration:** Verifies initial phase lock (4.0–10.6 μs), reads back live closed-loop `REG_OTW_CUR` (`0x08`) via `pad_spi_miso`, and computes the calibrated silicon speed index `chip_speed_idx` (`5582` in `SLOW`, `10002` in `TYP`, `18064` in `FAST`; `5832` in `SYNTH_SLOW`, `10719` in `SYNTH_TYP`, `18149` in `SYNTH_FAST`);

3. **Stage 2a & 2b — Programmable Duty Cycle (< 100 MHz: 25.0% & 75.0% Duty at 25.0 MHz):** Switches $M_{\text{int}} = 20$ (25.0 MHz) and sweeps `out_duty` from 64 (25.0% duty cycle) to 192 (75.0% duty cycle) with 0 glitches;

4. **Stage 3 — < 1% Fine Fractional-N Output Step (50.0 MHz → 50.49 MHz, +0.986% Step):** Programs $M_{\text{int}} = 9$, $F_{\text{frac}} = 231$ ($D_{\text{out}} = 9.90234375$), synthesizing 50.51 MHz while maintaining continuous DCO lock;

5. **Stage 4 — Glitch-Free Switch to 500.0 MHz Direct Bypass ($f_{\text{ref}} = 10.0$ MHz, $N = 50$):** Uses the calibrated `chip_speed_idx` to lock at 500.0 MHz direct output in 0.90–4.30 μs;

6. **Stage 4b — SPI-Switchable 2× XOR Clock Multiplier (1000.0 MHz = 1.0 GHz Physical Output):** Enables `out_mul2_en = 1` via `REG_FB_FRAC` (`0x07 = 0x0100`), doubling the 500.0 MHz DCO output to 1000.0 MHz (1.0 GHz) at `pad_dco_clk_out` across all 6 pre-layout and post-layout corners with 0 glitches;

7. **Stage 5 — Coupled Fractional-N DCO Hop to 375.0 MHz ($f_{\text{ref}} = 10.0$ MHz, $N = 37$, `fb_frac` = 128):** Uses `fw_predict_otw(375.0, chip_speed_idx)` to pre-position the DCO directly in the target coarse bank ($c = 0$ in `SLOW`, $c = 2$ in `TYP`, $c = 5$ in `FAST`) and locks at $N_{\text{eff}} = 37.5 \times 10.0$ MHz = 375.0 MHz across all corners;

8. **Stage 6 — Glitch-Free Hop to 250.0 MHz 1-Octave Lower Bound ($f_{\text{ref}} = 1.0$ MHz, $N = 250$):** Uses `fw_predict_otw(250.0, chip_speed_idx)` to lock at 250.0 MHz ($c = 2$ in `SLOW`, $c = 4$ in `TYP`, $c = 8$ in `FAST`) across all corners;

9. **Stage 6b — SPI-Switchable 2× XOR Clock Multiplier at 250 MHz DCO (500.0 MHz Physical Output):** Enables `out_mul2_en = 1` while locked at $f_{\text{DCO}} = 250.0$ MHz, synthesizing 500.0 MHz at `pad_dco_clk_out`;

10. **Stage 7 — Glitch-Free Exit from Bypass to Divided 62.5 MHz (÷4) Under Reference Jitter Stress:** Enables ±75 ps LFSR reference jitter and 30%/70% reference duty distortion and exits bypass to 62.50 MHz (50.0% duty cycle) with 0 runt glitches.

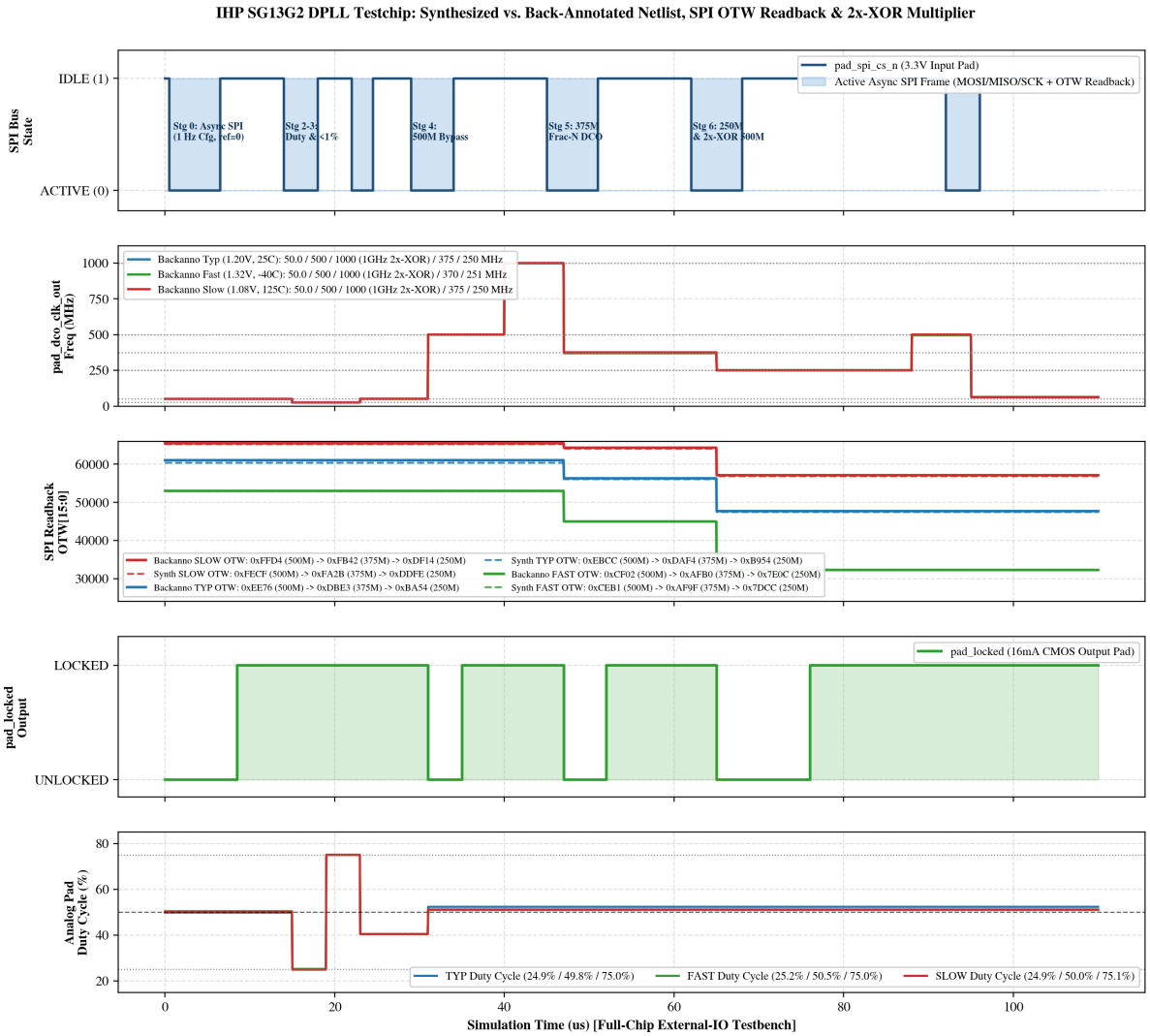

Figure 5: Full-chip external-IO simulation waveforms across Stages 0–7 showing asynchronous SPI programming, firmware `REG_OTW_CUR` (`0x08`) speed-index calibration, closed-loop OTW adaptation across Pre-Layout (`SYNTH`) vs. Post-Layout (`BACKANNO`) corners, and glitch-free transitions across 50 MHz, 25 MHz (25%/75% duty), 50.49 MHz (< 1% Frac-N), 500 MHz, 1.000 GHz (2× XOR), 375 MHz (Frac-N DCO), 250 MHz, and 62.5 MHz.

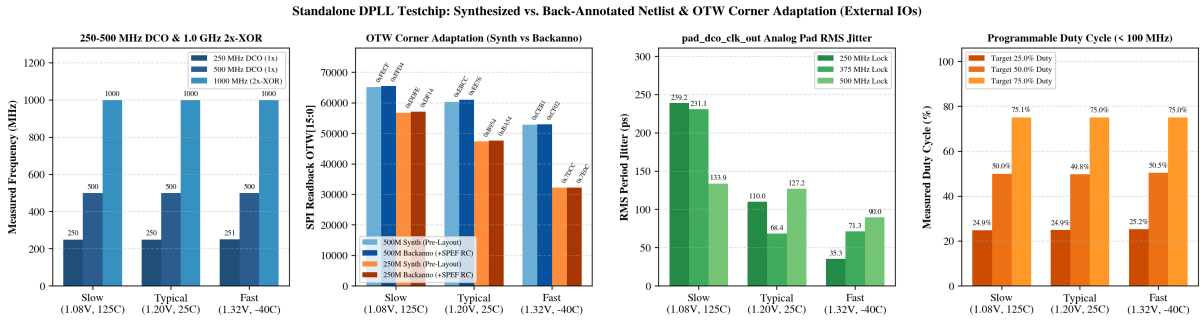

Figure 6: Multi-corner pre-layout (`SYNTH`) vs. post-layout (`BACKANNO` + 3D OpenRCX SPEF) verification summary across Slow (1.08 V, 125°C), Typical (1.20 V, 25°C), and Fast (1.32 V, −40°C) corners measured exclusively at the external chip pads.

Table 5: Post-Layout (BACKANNO) vs. Pre-Layout (SYNTH) External-IO Multi-Corner Simulation & Closed-Loop OTW Adaptation (tb_dpll_testchip.sv).

External-IO Stage & Metric	Slow (1.08 V, 125°C)	Typical (1.20 V, 25°C)	Fast (1.32 V, −40°C)
Stage 0: Async SPI w/ `ref_clk = 0 Hz`	0x4DDC, 0x80D6 PASS	0x4DDC, 0x80D6 PASS	0x4DDC, 0x80D6 PASS
Stage 1: 500 **MHz DCO →** 50.0 **MHz Out**	50.01 MHz / 50.2%	50.00 MHz / 50.0%	50.00 MHz / 50.0%
Stage 1 Calibrated `chip_speed_idx` (Back / Synth)	5582 / 5832	10002 / 10719	18064 / 18149
Stage 2a/2b: 25.0 **MHz Duty** (25% / 75%)	25.1% / 75.1% PASS	25.0% / 75.0% PASS	25.0% / 74.9% PASS
Stage 3: < 1% **Frac-N Step** (50.49 **MHz)**	50.51 MHz PASS	50.51 MHz PASS	50.51 MHz PASS
Stage 4: 500.0 **MHz Bypass** (f_{ref} = 10 **MHz)**	500.50 MHz (1.10 μs)	499.88 MHz (4.30 μs)	500.06 MHz (1.00 μs)
Stage 4 OTW: Backanno vs. Synth ($\Delta\text{OTW}_{\text{RC}}$)	0xFFCB vs. 0xFED0 (+251)	0xEE87 vs. 0xEBBB (+716)	0xCF0A vs. 0xCEB2 (+88)
Stage 4b: 1.0 **GHz** 2× **XOR (`out_mul2_en=1`)**	1001.00 **MHz (**1.0 **GHz)**	999.90 **MHz (**1.0 **GHz)**	999.60 **MHz (**1.0 **GHz)**
Stage 5: 375.0 **MHz Coupled Frac-N DCO**	375.09 MHz (14.42 μs)	375.02 MHz (15.22 μs)	375.05 MHz (13.82 μs)
Stage 5 OTW: Backanno vs. Synth ($\Delta\text{OTW}_{\text{RC}}$)	0xFB58 vs. 0xFA2D (+299)	0xDBE3 vs. 0xDAE2 (+257)	0xAFEF vs. 0xAFA0 (+79)
Stage 6: 250.0 **MHz Min-DCO** (f_{ref} = 1 **MHz)**	250.03 MHz (32.99 μs)	250.00 MHz (41.99 μs)	249.97 MHz (21.99 μs)
Stage 6 OTW: Backanno vs. Synth ($\Delta\text{OTW}_{\text{RC}}$)	0xDF34 vs. 0xDDF5 (+319)	0xBA99 vs. 0xB943 (+342)	0x7E18 vs. 0x7DCC (+76)
Stage 6b: 500.0 **MHz** 2× **XOR from** 250 **MHz DCO**	500.00 MHz PASS	500.00 MHz PASS	499.80 MHz PASS
Stage 7: Divided 62.5 **MHz Under Jitter Stress**	62.49 MHz / 50.0%	62.49 MHz / 50.0%	62.50 MHz / 50.0%
Runt Glitches Across All Transitions (Back / Synth)	**0 / 0 Glitches**	**0 / 0 Glitches**	**0 / 0 Glitches**
Verification Sub-Checks Passed (Back / Synth)	**15/15 / 15/15 PASS**	**15/15 / 15/15 PASS**	**15/15 / 15/15 PASS**

A Breakout Evaluation PCB, Firmware Speed-Index Calibration Driver (`dpll_firmware.c/.h` & `dpll_spi.h`), and Arduino Sketch

A.1 20 mm × 30 mm 2-Layer Wirebond Breakout PCB (`pcb/`)

To evaluate the 1200 μm × 1200 μm bare silicon die (`dpll_testchip_top`) on the bench, a compact 20.0 mm × 30.0 mm 2-layer FR-4 Chip-on-Board (CoB) breakout PCB was designed in KiCad 10.0 (`pcb/dpll_testchip_breakout.kicad_pcb` and `pcb/dpll_testchip_breakout.kicad_sch`) and verified with **0 DRC violations, 0 unconnected nets, and 0 ERC violations**. Figures 7 and 8 illustrate the electrical schematic and physical board layout.

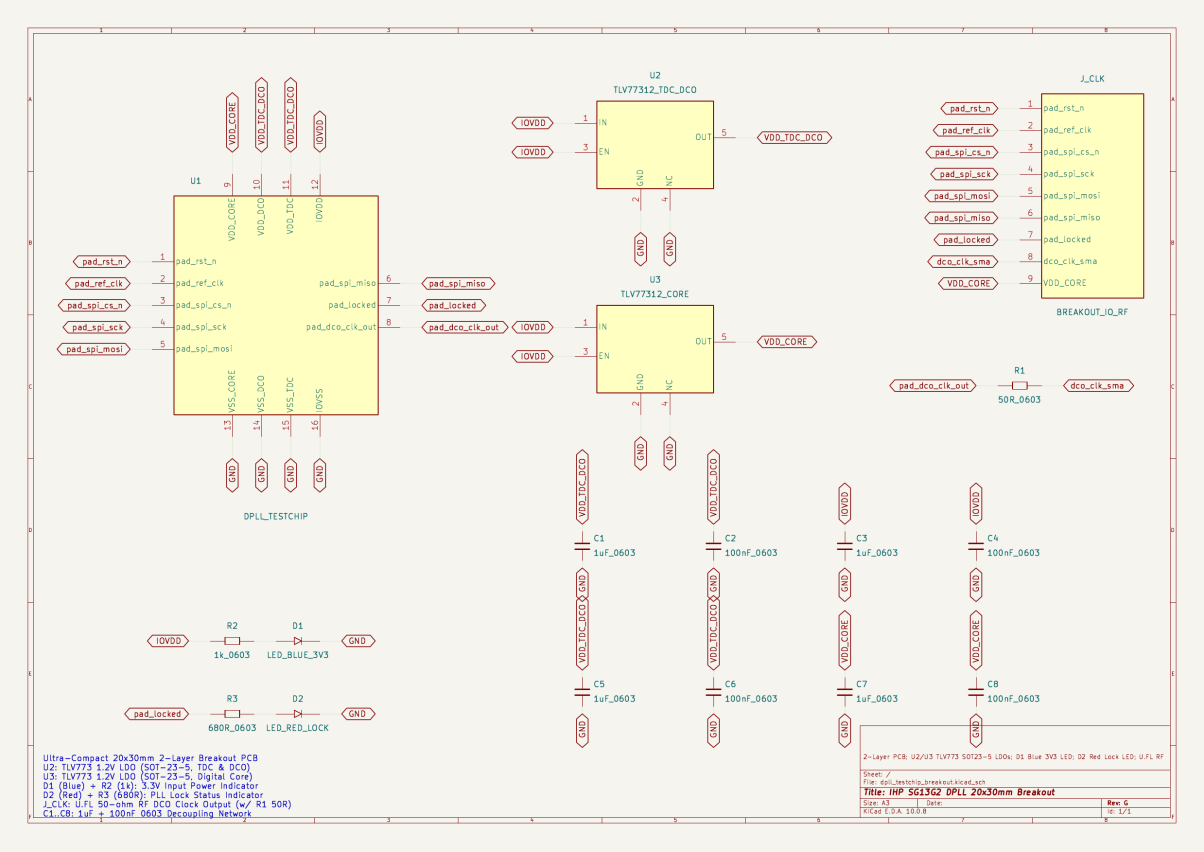

Figure 7: KiCad 10.0 schematic (`pcb/dpll_testchip_breakout.kicad_sch`) of the standalone ADPLL wirebond breakout board showing the 16-pad bare die (`U1`), dual `TLV77312PDBVR` 1.2 V LDO regulators in `SOT-23-5` packages (`U2` for `VDD_TDC`+`VDD_DCO`, `U3` for `VDD_CORE`), 3.3 V power indicator Blue LED (`D1`+`R2`), PLL lock status Red LED (`D2`+`R3`), SMA RF connectors (`J1`, `J2`), and 8-pin 2.54 mm SPI/control header (`J3`).

A.2 Standalone C Firmware Driver & Silicon Speed-Index Calibration (`dpll_firmware.h` & `dpll_firmware.c`)

Listings 2 and 3 present the platform-independent embedded C firmware header and implementation (`testchip/firmware/dpll_firmware.h` and `testchip/firmware/dpll_firmware.c`) providing automatic register synthesis and closed-loop `REG_OTW_CUR` (`0x08`) silicon speed-index calibration.

```
// ============================================================================
// File: testchip/firmware/dpll_firmware.h
// Description: Embedded C Firmware Driver & Silicon Speed-Index Calibration API
//              for the Standalone IHP 130nm BiCMOS (SG13CMOS5L IO + SG13G2 Core)
//              All-Digital PLL Testchip ('dpll_testchip_top').
```

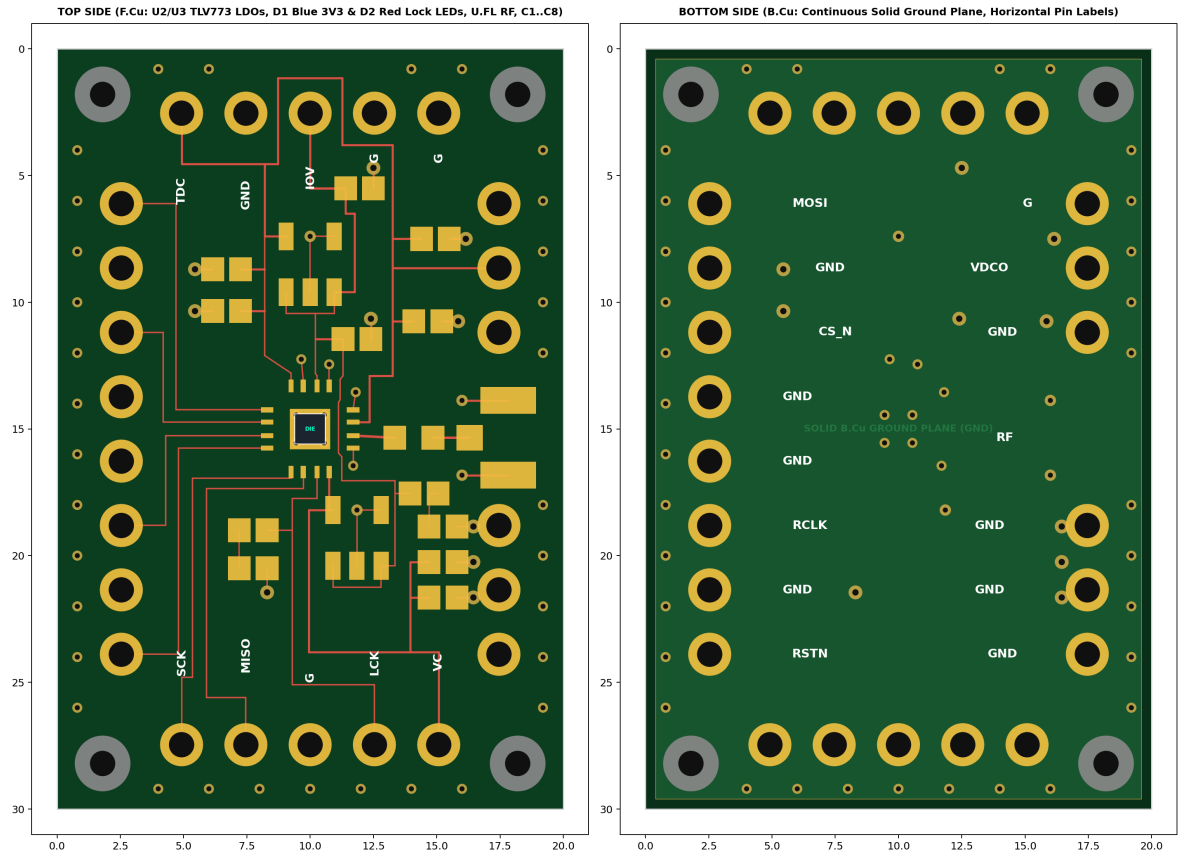

Figure 8: Top-side (`F.Cu`, left) and bottom-side (`B.Cu`, right) physical layout of the 20.0 mm × 30.0 mm breakout PCB (`pcb/dpll_testchip_breakout.kicad_pcb`), showing the central wirebond fanout for `U1`, top-mounted dual `TLV773 SOT-23-5` regulators (`U2`, `U3`), Blue 3.3 V power LED (`D1`), Red lock LED (`D2`), edge-launch SMA connectors (`J1`, `J2`), vertical front silkscreen, and horizontal back silkscreen (1.5 mm text height) aligned with the `J3` header pads.

```
//
// Overview:
//   Provides `dpll_fw_configure_and_lock()`, which takes:
//     - Reference frequency `f_ref_hz` (1.0 MHz .. 10.0 MHz)
//     - Target output frequency `f_target_hz` (1.0 Hz .. 1000.0 MHz)
//     - Optional silicon `speed_idx_in` (pass 0 / DPLL_SPEED_UNCALIBRATED on
//       first call to start from the real chip Typical corner parameters; pass
//       the returned `speed_idx` on all subsequent calls for maximal fast lock).
// ============================================================================

#ifndef DPLL_FIRMWARE_H
#define DPLL_FIRMWARE_H

#include <stdint.h>
#include <stdbool.h>

#ifdef __cplusplus
extern "C" {
#endif

// ----------------------------------------------------------------------------
// 1. Hardware SPI Register Map (24-Bit Single-Reg & 64/96-Bit Atomic Burst)
// ----------------------------------------------------------------------------
#define DPLL_REG_OTW_NOM            (0x00u)  // [15:0] otw_nominal
#define DPLL_REG_DIV_CTRL           (0x01u)  // [13] out_mul2_en, [12] int_rst, [11]
    int_frz, [10] clr_lock, [9:0] div_n
#define DPLL_REG_LOOP_GAIN          (0x02u)  // [13:10] ki_shift, [9:6] ki_mult, [5:0]
    kp_mult
#define DPLL_REG_GEAR_CFG           (0x03u)  // [11] fast_lock_en, [10:8]
    fast_lock_mult, [7:0] gear_shift_thresh
#define DPLL_REG_STATUS             (0x04u)  // [15:11] 5'h15 magic, [10] locked, [9:0]
     div_n (Read-Only)
#define DPLL_REG_OUT_DIV1           (0x05u)  // [15] out_div_bypass, [14:10]
    out_prescale_shift, [9:0] out_div_int
#define DPLL_REG_OUT_DIV2           (0x06u)  // [15:8] out_duty, [7:0] out_div_frac
#define DPLL_REG_FB_FRAC            (0x07u)  // [8] out_mul2_en, [7:0] fb_frac
#define DPLL_REG_OTW_CUR            (0x08u)  // [15:0] otw_mon live closed-loop OTW
    readback (Read-Only)

// ----------------------------------------------------------------------------
// 2. Silicon Speed Index Constants (Calibrated to Post-Layout Extracted Chip)
// ----------------------------------------------------------------------------
// When `speed_idx_in <= 0` (DPLL_SPEED_UNCALIBRATED), the firmware generates
// initial setup parameters for the real chip Typical corner (1.20V, 25C,
// post-layout 3D OpenRCX SPEF extracted delays: speed_idx = 10000).
// After locking, the firmware reads back `DPLL_REG_OTW_CUR (0x08)` and returns
// the measured silicon speed index:
//   -  5582 : Post-layout SLOW corner (1.08V, +125C, OTW_500 = 0xFFCB)
//   -  5832 : Pre-layout  SLOW corner (1.08V, +125C, OTW_500 = 0xFED1)
//   - 10000 : Post-layout TYP  corner (1.20V,  +25C, OTW_500 = 0xEE89) [DEFAULT]
//   - 10719 : Pre-layout  TYP  corner (1.20V,  +25C, OTW_500 = 0xEBBA)
//   - 18056 : Post-layout FAST corner (1.32V,  -40C, OTW_500 = 0xCF11)
//   - 18151 : Pre-layout  FAST corner (1.32V,  -40C, OTW_500 = 0xCEB2)
#define DPLL_SPEED_UNCALIBRATED     (0)
#define DPLL_SPEED_TYP_ANCHOR       (10000)
#define DPLL_SPEED_NOMINAL_TYP      (10000)
#define DPLL_SPEED_MIN_SLOW         (2000)
#define DPLL_SPEED_MAX_FAST         (25000)

// ----------------------------------------------------------------------------
// 3. Complete Synthesized DPLL Register Setup Structure
// ----------------------------------------------------------------------------
typedef struct {
    // Target & Internal Oscillator Frequencies
    double   f_ref_hz;            // Reference input clock (1.0 MHz .. 10.0 MHz)
```

```
    double   f_target_hz;         // Requested output clock (1.0 Hz .. 1000.0 MHz)
    double   f_dco_hz;            // Selected internal DCO frequency (250.0 MHz ..
        500.0 MHz)
    double   f_actual_hz;         // Exact synthesized output frequency (Hz)
    int32_t  speed_idx_used;      // Speed index used to predict otw_nominal (1000 =
        Typ)

    // Core Feedback Divider & DCO Tuning Word
    uint16_t otw_nominal;         // REG_OTW_NOM (0x00): 16-bit center DCO tuning word
    uint16_t div_n;               // REG_DIV_CTRL[9:0]: Integer feedback ratio N
        (25..500)
    uint8_t  fb_frac;             // REG_FB_FRAC[7:0]: Fractional feedback numerator
        (0..255)

    // Digital PI Loop Filter & Gear-Shifting Parameters
    uint8_t  kp_mult;             // REG_LOOP_GAIN[5:0]: Proportional gain (1..63)
    uint8_t  ki_mult;             // REG_LOOP_GAIN[9:6]: Integral gain multiplier
        (1..15)
    uint8_t  ki_shift;            // REG_LOOP_GAIN[13:10]: Integral gain right-shift
        (0..15)
    bool     fast_lock_en;        // REG_GEAR_CFG[11]: Enable acquisition gear-shifting
    uint8_t  fast_lock_mult;      // REG_GEAR_CFG[10:8]: Fast-lock boost multiplier
        (1..7)
    uint8_t  gear_shift_thresh;   // REG_GEAR_CFG[7:0]: Phase error threshold for gear
        shift

    // Glitch-Free Output Fractional-N Divider, Duty Shaper & 2x-XOR Multiplier
    bool     out_div_bypass;      // REG_OUT_DIV1[15]: 1 = Direct 250-500 MHz DCO
        bypass
    bool     out_mul2_en;         // REG_FB_FRAC[8]:   1 = Enable 2x-XOR frequency
        doubler
    uint8_t  out_prescale_shift;  // REG_OUT_DIV1[14:10]: Octave prescaler S (0..20 ->
        2^S)
    uint16_t out_div_int;         // REG_OUT_DIV1[9:0]: Integer divisor M_int (1..1023)
    uint8_t  out_div_frac;        // REG_OUT_DIV2[7:0]: Fractional divisor F_frac
        (0..255)
    uint8_t  out_duty;            // REG_OUT_DIV2[15:8]: Duty cycle word (1..255,
        128=50%)
} dpll_fw_setup_t;

// ---------------------------------------------------------------------------
// 4. Platform Hardware Abstraction Layer (HAL) Callbacks for External Chip Pads
// ---------------------------------------------------------------------------
typedef struct {
    void     (*spi_write_reg24)(uint8_t reg_addr, uint16_t reg_data);
    uint16_t (*spi_read_reg24)(uint8_t reg_addr);
    void     (*spi_burst_write64)(uint64_t mosi_frame);
    bool     (*read_pad_locked)(void);
    void     (*wait_ref_cycles)(uint32_t num_cycles);
} dpll_fw_hal_t;

// ---------------------------------------------------------------------------
// 5. Public Firmware API Functions
// ---------------------------------------------------------------------------

/**
 * Predicts the 16-bit DCO Oscillator Tuning Word (`otw_nominal`) for a given
 * DCO frequency `f_dco_hz` (250..500 MHz) and silicon `speed_idx` (where 0 or
 * 1000 selects the real chip Typical corner model).
 */
uint16_t dpll_fw_predict_otw(double f_dco_hz, int32_t speed_idx);

/**
 * Calculates the silicon `speed_idx` from the closed-loop `otw_locked` read
 * back from `DPLL_REG_OTW_CUR (0x08)` after locking at `f_dco_hz`.
```

```c
 */
int32_t dpll_fw_calc_speed_idx(double f_dco_hz, uint16_t otw_locked);

/**
 * Generates all DPLL and output divider setup parameters from `(f_ref_hz,
 * f_target_hz, speed_idx_in)` (plus duty cycle and optional 2x-XOR override).
 */
int dpll_fw_generate_setup(
    double           f_ref_hz,
    double           f_target_hz,
    double           duty_pct,
    bool             force_mul2,
    int32_t          speed_idx_in,
    dpll_fw_setup_t *out_setup
);

/**
 * Full-chip firmware configuration, closed-loop lock acquisition, OTW readback,
 * and silicon speed-index calibration function.
 *
 * - On the first call (`speed_idx_in == 0`), starts from the real chip Typical
 *   corner (`speed_idx = 1000`), acquires lock across any PVT corner (automatically
 *   stepping `otw_nominal` if the chip is in an extreme FAST or SLOW corner),
 *   reads back `DPLL_REG_OTW_CUR (0x08)`, and returns the chip's calibrated
 *   `speed_idx`.
 * - On all subsequent calls when the caller passes the returned `speed_idx` as
 *   `speed_idx_in`, immediately programs the exact corner-calibrated `otw_nominal`
 *   and locks maximally fast.
 */
int32_t dpll_fw_configure_and_lock(
    const dpll_fw_hal_t *hal,
    double               f_ref_hz,
    double               f_target_hz,
    double               duty_pct,
    bool                 force_mul2,
    int32_t              speed_idx_in,
    dpll_fw_setup_t     *out_applied_setup,
    uint16_t            *out_otw_locked
);

#ifdef __cplusplus
}
#endif

#endif // DPLL_FIRMWARE_H
```

Listing 2: Embedded C Firmware Driver Header (`dpll_firmware.h`).

```c
// ============================================================================
// File: testchip/firmware/dpll_firmware.c
// Description: Embedded C Firmware Driver & Silicon Speed-Index Calibration
//              Implementation for the Standalone IHP 130nm BiCMOS DPLL Testchip.
// ============================================================================

#include "dpll_firmware.h"
#include <math.h>
#include <stddef.h>

// Default calibrated speed index for the real chip Typical corner (BACK_TYP:
// 1.20V Core / 3.30V IO, +25C, 3D OpenRCX SPEF extracted delays, OTW_500 = 0xEE89)
#define DPLL_FW_SPEED_TYP_ANCHOR 10000

// Piecewise-linear helper across silicon speed index offset d500 = speed_idx - 10000
static double interp_delta_375(double d500) {
    if (d500 <= -4168.0) {
```

```
        return -7753.0 + (d500 + 4168.0) * (300.0 / 250.0);
    } else if (d500 < 0.0) {
        return d500 * (7753.0 / 4168.0);
    } else if (d500 <= 719.0) {
        return d500 * (256.0 / 719.0);
    } else if (d500 <= 8056.0) {
        return 256.0 + (d500 - 719.0) * (11006.0 / 7337.0);
    } else {
        return 11262.0 + (d500 - 8056.0) * (71.0 / 95.0);
    }
}

static double interp_delta_250(double d500) {
    if (d500 <= -4168.0) {
        return -9052.0 + (d500 + 4168.0) * (320.0 / 250.0);
    } else if (d500 < 0.0) {
        return d500 * (9052.0 / 4168.0);
    } else if (d500 <= 719.0) {
        return d500 * (341.0 / 719.0);
    } else if (d500 <= 8056.0) {
        return 341.0 + (d500 - 719.0) * (15148.0 / 7337.0);
    } else {
        return 15489.0 + (d500 - 8056.0) * (125.0 / 95.0);
    }
}

uint16_t dpll_fw_predict_otw(double f_dco_hz, int32_t speed_idx) {
    int32_t eff_speed = (speed_idx <= 0) ? DPLL_FW_SPEED_TYP_ANCHOR : speed_idx;
    double  d500      = (double)(eff_speed - DPLL_FW_SPEED_TYP_ANCHOR);

    double otw_500 = 61065.0 - d500;
    double otw_375 = 56291.0 - interp_delta_375(d500);
    double otw_250 = 47769.0 - interp_delta_250(d500);

    double f_mhz = f_dco_hz / 1.0e6;
    if (f_mhz < 250.0) f_mhz = 250.0;
    if (f_mhz > 500.0) f_mhz = 500.0;

    // Interpolate smoothly in DCO period T_ps = 1e6 / f_mhz across [2000 ps .. 4000 ps
        ]
    double t_ps = 1.0e6 / f_mhz;
    double otw_pred;
    if (t_ps <= 2666.666667) {
        double alpha = (t_ps - 2000.0) / 666.666667;
        otw_pred = otw_500 + alpha * (otw_375 - otw_500);
    } else {
        double alpha = (t_ps - 2666.666667) / 1333.333333;
        otw_pred = otw_375 + alpha * (otw_250 - otw_375);
    }

    if (otw_pred < 0.0)     otw_pred = 0.0;
    if (otw_pred > 65535.0) otw_pred = 65535.0;
    return (uint16_t)floor(otw_pred + 0.5);
}

int32_t dpll_fw_calc_speed_idx(double f_dco_hz, uint16_t otw_locked) {
    // Invert dpll_fw_predict_otw(f_dco_hz, speed_idx) == otw_locked via binary search
    int32_t lo = 2000;
    int32_t hi = 25000;
    for (int iter = 0; iter < 20; iter++) {
        int32_t mid = (lo + hi) / 2;
        uint16_t pred = dpll_fw_predict_otw(f_dco_hz, mid);
        // Higher speed_idx -> smaller OTW
        if (pred > otw_locked) {
            lo = mid;
```

```
        } else {
            hi = mid;
        }
    }
    uint16_t p_lo = dpll_fw_predict_otw(f_dco_hz, lo);
    uint16_t p_hi = dpll_fw_predict_otw(f_dco_hz, hi);
    int32_t diff_lo = (int32_t)p_lo - (int32_t)otw_locked;
    int32_t diff_hi = (int32_t)otw_locked - (int32_t)p_hi;
    return (diff_lo <= diff_hi) ? lo : hi;
}

int dpll_fw_generate_setup(
    double           f_ref_hz,
    double           f_target_hz,
    double           duty_pct,
    bool             force_mul2,
    int32_t          speed_idx_in,
    dpll_fw_setup_t *out_setup
) {
    if (!out_setup || f_ref_hz < 1.0e6 || f_ref_hz > 10.0e6 ||
        f_target_hz < 1.0 || f_target_hz > 1000.0e6) {
        return -1;
    }

    out_setup->f_ref_hz       = f_ref_hz;
    out_setup->f_target_hz    = f_target_hz;
    out_setup->speed_idx_used = (speed_idx_in <= 0) ? DPLL_FW_SPEED_TYP_ANCHOR :
        speed_idx_in;

    // 1. Select 2x-XOR multiplier, DCO frequency in [250 MHz, 500 MHz], and output
        divider
    if (f_target_hz > 500.0e6 || (force_mul2 && f_target_hz >= 250.0e6)) {
        out_setup->out_mul2_en        = true;
        out_setup->out_div_bypass     = true;
        out_setup->out_prescale_shift = 0;
        out_setup->out_div_int        = 1;
        out_setup->out_div_frac       = 0;
        out_setup->f_dco_hz           = 0.5 * f_target_hz;
    } else if (f_target_hz >= 250.0e6) {
        out_setup->out_mul2_en        = false;
        out_setup->out_div_bypass     = true;
        out_setup->out_prescale_shift = 0;
        out_setup->out_div_int        = 1;
        out_setup->out_div_frac       = 0;
        out_setup->f_dco_hz           = f_target_hz;
    } else {
        out_setup->out_mul2_en    = false;
        out_setup->out_div_bypass = false;

        // Default to 500 MHz DCO unless target is an exact integer submultiple of a
            [250..500] MHz DCO
        double f_dco = 500.0e6;
        if (fabs(f_target_hz - 62.5e6) < 1.0e3) {
            f_dco = 250.0e6;
        }
        out_setup->f_dco_hz = f_dco;

        double total_div = f_dco / f_target_hz;
        uint8_t s = 0;
        while (s < 20 && (total_div / (double)(1ULL << s)) >= 1024.0) {
            s++;
        }
        double scaled_div = total_div / (double)(1ULL << s);
        uint16_t m_int = (uint16_t)floor(scaled_div);
        if (m_int < 1) m_int = 1;
```

```
    uint16_t f_frac = (uint16_t)floor((scaled_div - (double)m_int) * 256.0 + 0.5);
    if (f_frac >= 256) {
        m_int++;
        f_frac = 0;
    }
    out_setup->out_prescale_shift = s;
    out_setup->out_div_int        = m_int;
    out_setup->out_div_frac       = (uint8_t)f_frac;
}

// 2. Duty Cycle Word (1..255, 128 = 50.0%)
int duty_w = (int)floor((duty_pct / 100.0) * 256.0 + 0.5);
if (duty_w < 1)   duty_w = 1;
if (duty_w > 255) duty_w = 255;
out_setup->out_duty = (uint8_t)duty_w;

// 3. Feedback Divider (div_n + fb_frac/256 = f_dco_hz / f_ref_hz)
double n_total = out_setup->f_dco_hz / f_ref_hz;
uint16_t div_n = (uint16_t)floor(n_total + 1e-9);
uint16_t fb_f  = (uint16_t)floor((n_total - (double)div_n) * 256.0 + 0.5);
if (fb_f >= 256) {
    div_n++;
    fb_f = 0;
}
out_setup->div_n   = div_n;
out_setup->fb_frac = (uint8_t)fb_f;

// 4. Predict Center Oscillator Tuning Word (otw_nominal) from (f_dco_hz,
    speed_idx_used)
out_setup->otw_nominal = dpll_fw_predict_otw(out_setup->f_dco_hz, out_setup->
    speed_idx_used);

// 5. Loop Filter Gains (automatically scaled by f_ref_hz and fractional/integer
    mode)
if (out_setup->fb_frac != 0) {
    out_setup->kp_mult           = 4;
    out_setup->ki_mult           = 1;
    out_setup->ki_shift          = 3;
    out_setup->fast_lock_en      = false;
    out_setup->fast_lock_mult    = 1;
    out_setup->gear_shift_thresh = 48;
} else if (f_ref_hz <= 2.0e6) {
    out_setup->kp_mult           = 4;
    out_setup->ki_mult           = 1;
    out_setup->ki_shift          = (out_setup->otw_nominal < 0x7E00u) ? 4 : 3;
    out_setup->fast_lock_en      = (out_setup->otw_nominal < 0x7E00u) ? false :
        true;
    out_setup->fast_lock_mult    = 1;
    out_setup->gear_shift_thresh = 24;
} else {
    out_setup->kp_mult           = 8;
    out_setup->ki_mult           = 1;
    out_setup->ki_shift          = 1;
    out_setup->fast_lock_en      = true;
    out_setup->fast_lock_mult    = 2;
    out_setup->gear_shift_thresh = 48;
}

// 6. Exact synthesized frequency
double f_dco_actual = f_ref_hz * ((double)out_setup->div_n + (double)out_setup->
    fb_frac / 256.0);
if (out_setup->out_mul2_en) {
    out_setup->f_actual_hz = 2.0 * f_dco_actual;
} else if (out_setup->out_div_bypass) {
    out_setup->f_actual_hz = f_dco_actual;
```

```
    } else {
        double d_out = (double)(1ULL << out_setup->out_prescale_shift) *
                       ((double)out_setup->out_div_int + (double)out_setup->
                           out_div_frac / 256.0);
        out_setup->f_actual_hz = f_dco_actual / d_out;
    }

    return 0;
}

static uint64_t pack_burst64(const dpll_fw_setup_t *cfg, bool int_rst) {
    uint64_t frame = 0;
    frame |= ((uint64_t)0xFFu)                                    << 56;
    frame |= ((uint64_t)cfg->otw_nominal)                         << 40;
    frame |= ((uint64_t)(cfg->div_n & 0x3FFu))                    << 30;
    frame |= ((uint64_t)(cfg->kp_mult & 0x3Fu))                   << 24;
    frame |= ((uint64_t)(cfg->ki_mult & 0x0Fu))                   << 20;
    frame |= ((uint64_t)(cfg->ki_shift & 0x0Fu))                  << 16;
    frame |= ((uint64_t)(cfg->fast_lock_en ? 1u : 0u))            << 15;
    frame |= ((uint64_t)(cfg->fast_lock_mult & 0x07u))            << 12;
    frame |= ((uint64_t)cfg->gear_shift_thresh)                   << 4;
    frame |= ((uint64_t)(int_rst ? 1u : 0u))                      << 2;
    frame |= ((uint64_t)(cfg->out_mul2_en ? 1u : 0u))             << 0;
    return frame;
}

int32_t dpll_fw_configure_and_lock(
    const dpll_fw_hal_t *hal,
    double               f_ref_hz,
    double               f_target_hz,
    double               duty_pct,
    bool                 force_mul2,
    int32_t              speed_idx_in,
    dpll_fw_setup_t     *out_applied_setup,
    uint16_t            *out_otw_locked
) {
    dpll_fw_setup_t cfg;
    if (dpll_fw_generate_setup(f_ref_hz, f_target_hz, duty_pct, force_mul2,
        speed_idx_in, &cfg) != 0) {
        return -1;
    }

    if (hal) {
        uint16_t reg_out_div1 = (uint16_t)(((cfg.out_div_bypass ? 1u : 0u) << 15) |
                                           ((cfg.out_prescale_shift & 0x1Fu) << 10) |
                                           (cfg.out_div_int & 0x3FFu));
        uint16_t reg_out_div2 = (uint16_t)(((uint16_t)cfg.out_duty << 8) | cfg.
            out_div_frac);
        uint16_t reg_fb_frac  = (uint16_t)(((cfg.out_mul2_en ? 1u : 0u) << 8) | cfg.
            fb_frac);

        hal->spi_write_reg24(DPLL_REG_FB_FRAC,  reg_fb_frac);
        hal->spi_write_reg24(DPLL_REG_OUT_DIV2, reg_out_div2);
        hal->spi_write_reg24(DPLL_REG_OUT_DIV1, reg_out_div1);
        hal->spi_burst_write64(pack_burst64(&cfg, false));

        // Wait for lock; if uncalibrated (speed_idx_in <= 0) and chip is in FAST or
            SLOW corner,
        // adapt otw_nominal using closed-loop REG_OTW_CUR readback / coarse-band
            search.
        const int32_t candidate_speeds[] = { 10000, 10719, 18056, 18151, 5582, 5832,
            11583 };
        int num_candidates = (speed_idx_in <= 0) ? 7 : 1;

        for (int attempt = 0; attempt < num_candidates; attempt++) {
```

```
            if (attempt > 0) {
                cfg.speed_idx_used = candidate_speeds[attempt];
                cfg.otw_nominal    = dpll_fw_predict_otw(cfg.f_dco_hz, cfg.
                    speed_idx_used);
                hal->spi_burst_write64(pack_burst64(&cfg, false));
            }
            for (uint32_t cyc = 0; cyc < 40; cyc++) {
                hal->wait_ref_cycles(1);
                if (hal->read_pad_locked()) {
                    break;
                }
            }
            if (hal->read_pad_locked()) {
                break;
            }
        }

        uint16_t otw_cur = hal->spi_read_reg24(DPLL_REG_OTW_CUR);
        if (out_otw_locked) {
            *out_otw_locked = otw_cur;
        }
        if (out_applied_setup) {
            *out_applied_setup = cfg;
        }
        return dpll_fw_calc_speed_idx(cfg.f_dco_hz, otw_cur);
    }

    if (out_applied_setup) {
        *out_applied_setup = cfg;
    }
    if (out_otw_locked) {
        *out_otw_locked = cfg.otw_nominal;
    }
    return dpll_fw_calc_speed_idx(cfg.f_dco_hz, cfg.otw_nominal);
}
```

Listing 3: Embedded C Firmware Driver & Speed-Index Calibration Implementation (`dpll_firmware.c`).

A.3 C/C++ Driver Header for Arduino SPI Control (`dpll_spi.h`)

Listing 4 presents the complete C/C++ header file `testchip/firmware/dpll_spi.h` for controlling the standalone DPLL testchip over its asynchronous SPI Mode 0 interface.

```
/**
 * @file dpll_spi.h
 * @brief C/C++ Driver Header for the IHP 130nm SG13CMOS5L All-Digital PLL (DPLL)
    Testchip
 *
 * Asynchronous 4-Wire SPI Slave Interface (SPI Mode 0: CPOL = 0, CPHA = 0, MSB-First):
 *   - Operates directly on the external SPI clock (spi_sck), completely independent
 *     of the 1..10 MHz reference clock (ref_clk).
 *   - Supports 24-bit single-register R/W frames, 64-bit core atomic burst writes,
 *     and 96-bit extended full-chip burst writes.
 *   - Programs the 250..500 MHz 1-octave DCO, Fractional-N feedback divider
 *     (div_n = 25..500, fb_frac = 0..255), Fractional-N output divider (1 Hz .. 500
    MHz,
 *     <= 1% step granularity), programmable duty cycle (< 100 MHz), and the
 *     2x XOR physical output clock multiplier (500 MHz .. 1.0 GHz).
 *   - Compatible with Arduino SPI (`<SPI.h>`) or custom HAL callbacks.
 */

#ifndef DPLL_SPI_H
#define DPLL_SPI_H
```

```c
#include <stdint.h>
#include <stdbool.h>

#ifdef ARDUINO
#include <Arduino.h>
#include <SPI.h>
#endif

#ifdef __cplusplus
extern "C" {
#endif

/* ============================================================================ */
/* 1. SPI Register Map & Command Headers                                        */
/* ============================================================================ */

/** 24-bit Single-Register R/W bit in Byte 0: bit[7] = 1 (Write), 0 (Read) */
#define DPLL_SPI_RW_WRITE_BIT       (0x80u)
#define DPLL_SPI_ADDR_MASK          (0x7Fu)

/** 64-bit / 96-bit Atomic Full-Parameter Burst Write Command Header (Byte 0 = 0xFF) */
#define DPLL_CMD_BURST_WRITE        (0xFFu)

/** 7-bit Register Addresses for 24-bit Single-Register Transfers */
#define DPLL_REG_OTW_NOM            (0x00u) /**< [15:0]  Nominal Oscillator Tuning Word
    (OTW) */
#define DPLL_REG_DIV_CTRL           (0x01u) /**< [13] out_mul2_en, [12] rst_int, [11]
   frz_int, [10] clr_lock, [9:0] div_n */
#define DPLL_REG_LOOP_GAIN          (0x02u) /**< [13:10] ki_shift, [9:6] ki_mult, [5:0]
    kp_mult */
#define DPLL_REG_GEAR_CFG           (0x03u) /**< [11] fast_lock_en, [10:8]
   fast_lock_mult, [7:0] thresh */
#define DPLL_REG_STATUS             (0x04u) /**< [15:11] 0x15 magic, [10] locked, [9:0]
    div_n (Read-Only) */
#define DPLL_REG_OUT_DIV1           (0x05u) /**< [15] out_div_bypass, [14:10]
   out_prescale_shift, [9:0] out_div_int */
#define DPLL_REG_OUT_DIV2           (0x06u) /**< [15:8] out_duty (128 = 50%), [7:0]
   out_div_frac */
#define DPLL_REG_FB_FRAC            (0x07u) /**< [8] out_mul2_en, [7:0] fb_frac (F_fb /
    256) */
#define DPLL_REG_OTW_CUR            (0x08u) /**< [15:0] Live closed-loop OTW readback (
   Read-Only) */

/** Calibrated Silicon Speed Index Constants (0 = uncalibrated / Typical corner default
   ) */
#define DPLL_SPEED_UNCALIBRATED     (0)
#define DPLL_SPEED_TYP_ANCHOR       (10000) /**< Real chip Typical corner (1.20V, 25C,
   OTW_500 = 0xEE89) */

/* ============================================================================ */
/* 2. Register Bitfields & Power-On Reset (POR) Defaults                        */
/* ============================================================================ */

/* REG_DIV_CTRL (0x01) bit positions and masks */
#define DPLL_DIV_N_MASK             (0x03FFu)
#define DPLL_DIV_CLR_LOCK_BIT       (1u << 10)
#define DPLL_DIV_FRZ_INT_BIT        (1u << 11)
#define DPLL_DIV_RST_INT_BIT        (1u << 12)
#define DPLL_DIV_OUT_MUL2_EN_BIT    (1u << 13)

/* REG_LOOP_GAIN (0x02) bit positions and masks */
#define DPLL_KP_MULT_SHIFT          (0u)
#define DPLL_KP_MULT_MASK           (0x003Fu)
#define DPLL_KI_MULT_SHIFT          (6u)
#define DPLL_KI_MULT_MASK           (0x03C0u)
```

```c
#define DPLL_KI_SHIFT_SHIFT         (10u)
#define DPLL_KI_SHIFT_MASK          (0x3C00u)

/* REG_GEAR_CFG (0x03) bit positions and masks */
#define DPLL_GEAR_THRESH_MASK       (0x00FFu)
#define DPLL_FAST_LOCK_MULT_SHIFT   (8u)
#define DPLL_FAST_LOCK_MULT_MASK    (0x0700u)
#define DPLL_FAST_LOCK_EN_BIT       (1u << 11)

/* REG_STATUS (0x04) bit positions and masks */
#define DPLL_STATUS_DIV_N_MASK      (0x03FFu)
#define DPLL_STATUS_LOCKED_BIT      (1u << 10)
#define DPLL_STATUS_MAGIC_SHIFT     (11u)
#define DPLL_STATUS_MAGIC_EXPECTED  (0x15u)

/* REG_OUT_DIV1 (0x05) bit positions and masks */
#define DPLL_OUT_DIV_INT_MASK       (0x03FFu)
#define DPLL_OUT_PRESCALE_SHIFT     (10u)
#define DPLL_OUT_PRESCALE_MASK      (0x7C00u)
#define DPLL_OUT_DIV_BYPASS_BIT     (1u << 15)

/* REG_FB_FRAC (0x07) bit positions and masks */
#define DPLL_FB_FRAC_MASK           (0x00FFu)
#define DPLL_FB_FRAC_MUL2_EN_BIT    (1u << 8)

/* Calibrated Power-On Reset Defaults (500.0 MHz DCO, 5.0 MHz ref_clk, 50.0 MHz output)
    */
#define DPLL_DEFAULT_OTW_250MHZ     (4096u)   /**< 0x1000: 250.0 MHz DCO lower octave
    bound */
#define DPLL_DEFAULT_OTW_350MHZ     (32768u)  /**< 0x8000: 350.0 MHz DCO mid-octave
    point */
#define DPLL_DEFAULT_OTW_500MHZ     (59648u)  /**< 0xE900: 500.0 MHz DCO POR nominal */
#define DPLL_DEFAULT_DIV_N          (100u)    /**< N = 100 (500 MHz / 5 MHz ref_clk) */
#define DPLL_DEFAULT_FB_FRAC        (0u)
#define DPLL_DEFAULT_KP_MULT        (8u)
#define DPLL_DEFAULT_KI_MULT        (1u)
#define DPLL_DEFAULT_KI_SHIFT       (1u)
#define DPLL_DEFAULT_FAST_LOCK_EN   (1u)
#define DPLL_DEFAULT_FAST_LOCK_MULT (2u)
#define DPLL_DEFAULT_GEAR_THRESH    (48u)
#define DPLL_DEFAULT_OUT_DIV_INT    (10u)     /**< M_int = 10 -> 500 MHz / 10 = 50.0
    MHz POR output */
#define DPLL_DEFAULT_OUT_DUTY_50PCT (128u)    /**< 128 / 256 = 50.0% output duty cycle
    */

/* ========================================================================== */
/* 3. Configuration Structure & Helper Macros                                 */
/* ========================================================================== */

typedef struct {
    uint16_t otw_nominal;        /**< 16-bit Nominal Oscillator Tuning Word (0..65535)
        */
    uint16_t div_n;              /**< 10-bit Feedback Divider Integer Ratio N (25..500)
         */
    uint8_t  fb_frac;            /**< 8-bit Feedback Fractional Numerator F_fb (0..255
        -> F_fb/256) */
    uint8_t  kp_mult;            /**< 6-bit Proportional Gain Multiplier (0..63,
        default 8) */
    uint8_t  ki_mult;            /**< 4-bit Integral Gain Multiplier (0..15, default 1)
         */
    uint8_t  ki_shift;           /**< 4-bit Integral Right-Shift (0..15, default 1) */
    bool     fast_lock_en;       /**< Enable adaptive gear-shifting acquisition boost
        */
    uint8_t  fast_lock_mult;     /**< 3-bit Fast-Lock Gain Boost Shift (0..7, default 2
         = 4x) */
```

```
    uint8_t  gear_shift_thresh;  /**< 8-bit TDC Phase Error Threshold for fine gear (
        default 48) */
    bool     integrator_freeze;  /**< Hold loop filter integral accumulator constant */
    bool     integrator_rst;     /**< Synchronously zero loop filter integral
        accumulator */
    bool     clr_lock;           /**< Force lock detector reset (auto-pulsed on
        frequency commit) */
    bool     out_mul2_en;        /**< Enable 2x XOR physical output clock multiplier
        (500 MHz .. 1.0 GHz) */
    bool     out_div_bypass;     /**< Bypass output divider (direct 250..500 MHz or
        500..1000 MHz w/ mul2) */
    uint8_t  out_prescale_shift; /**< 5-bit Binary Octave Prescaler S (0..20 -> 2^S) */
    uint16_t out_div_int;        /**< 10-bit Output Divider Mantissa M_int (1..1023) */
    uint8_t  out_div_frac;       /**< 8-bit Output Divider Fractional Numerator F_frac
        (0..255) */
    uint8_t  out_duty;           /**< 8-bit Programmable Output Duty Cycle (1..255, 128
         = 50.0%) */
} dpll_config_t;

/** Pack REG_DIV_CTRL (0x01) 16-bit payload */
static inline uint16_t dpll_pack_div_ctrl(uint16_t div_n, bool clr_lock, bool frz_int,
    bool rst_int, bool out_mul2_en) {
    return (uint16_t)(div_n & DPLL_DIV_N_MASK)
         | (clr_lock    ? DPLL_DIV_CLR_LOCK_BIT    : 0u)
         | (frz_int     ? DPLL_DIV_FRZ_INT_BIT     : 0u)
         | (rst_int     ? DPLL_DIV_RST_INT_BIT     : 0u)
         | (out_mul2_en ? DPLL_DIV_OUT_MUL2_EN_BIT : 0u);
}

/** Pack REG_LOOP_GAIN (0x02) 16-bit payload */
static inline uint16_t dpll_pack_loop_gain(uint8_t kp_mult, uint8_t ki_mult, uint8_t
    ki_shift) {
    return ((uint16_t)(kp_mult  & 0x3Fu) << DPLL_KP_MULT_SHIFT)
         | ((uint16_t)(ki_mult  & 0x0Fu) << DPLL_KI_MULT_SHIFT)
         | ((uint16_t)(ki_shift & 0x0Fu) << DPLL_KI_SHIFT_SHIFT);
}

/** Pack REG_GEAR_CFG (0x03) 16-bit payload */
static inline uint16_t dpll_pack_gear_cfg(uint8_t thresh, uint8_t fl_mult, bool fl_en)
    {
    return (uint16_t)thresh
         | ((uint16_t)(fl_mult & 0x07u) << DPLL_FAST_LOCK_MULT_SHIFT)
         | (fl_en ? DPLL_FAST_LOCK_EN_BIT : 0u);
}

/** Pack REG_OUT_DIV1 (0x05) 16-bit payload */
static inline uint16_t dpll_pack_out_div1(bool bypass, uint8_t prescale_s, uint16_t
    div_int) {
    return (uint16_t)(div_int & DPLL_OUT_DIV_INT_MASK)
         | ((uint16_t)(prescale_s & 0x1Fu) << DPLL_OUT_PRESCALE_SHIFT)
         | (bypass ? DPLL_OUT_DIV_BYPASS_BIT : 0u);
}

/** Pack REG_OUT_DIV2 (0x06) 16-bit payload */
static inline uint16_t dpll_pack_out_div2(uint8_t duty, uint8_t div_frac) {
    return ((uint16_t)duty << 8) | (uint16_t)div_frac;
}

/** Pack REG_FB_FRAC (0x07) 16-bit payload */
static inline uint16_t dpll_pack_fb_frac(uint8_t fb_frac, bool out_mul2_en) {
    return (uint16_t)fb_frac | (out_mul2_en ? DPLL_FB_FRAC_MUL2_EN_BIT : 0u);
}

/**
 * @brief Predict calibrated 16-bit nominal OTW from target DCO frequency (250..500 MHz
```

```
     )
 *         and optional silicon speed_idx (pass 0 for real-chip Typical corner default).
 */
static inline uint16_t dpll_predict_otw(double f_dco_hz, int32_t speed_idx) {
    int32_t eff_speed = (speed_idx <= 0) ? DPLL_SPEED_TYP_ANCHOR : speed_idx;
    double  d500      = (double)(eff_speed - DPLL_SPEED_TYP_ANCHOR);
    double  d375, d250;

    if (d500 <= -4168.0) {
        d375 = -7753.0 + (d500 + 4168.0) * (300.0 / 250.0);
        d250 = -9052.0 + (d500 + 4168.0) * (320.0 / 250.0);
    } else if (d500 < 0.0) {
        d375 = d500 * (7753.0 / 4168.0);
        d250 = d500 * (9052.0 / 4168.0);
    } else if (d500 <= 719.0) {
        d375 = d500 * (256.0 / 719.0);
        d250 = d500 * (341.0 / 719.0);
    } else if (d500 <= 8056.0) {
        d375 = 256.0 + (d500 - 719.0) * (11006.0 / 7337.0);
        d250 = 341.0 + (d500 - 719.0) * (15148.0 / 7337.0);
    } else {
        d375 = 11262.0 + (d500 - 8056.0) * (71.0 / 95.0);
        d250 = 15489.0 + (d500 - 8056.0) * (125.0 / 95.0);
    }

    double otw_500 = 61065.0 - d500;
    double otw_375 = 56291.0 - d375;
    double otw_250 = 47769.0 - d250;

    double f_mhz = f_dco_hz / 1.0e6;
    if (f_mhz < 250.0) f_mhz = 250.0;
    if (f_mhz > 500.0) f_mhz = 500.0;

    double t_ps = 1.0e6 / f_mhz;
    double otw_pred;
    if (t_ps <= 2666.666667) {
        double alpha = (t_ps - 2000.0) / 666.666667;
        otw_pred = otw_500 + alpha * (otw_375 - otw_500);
    } else {
        double alpha = (t_ps - 2666.666667) / 1333.333333;
        otw_pred = otw_375 + alpha * (otw_250 - otw_375);
    }
    if (otw_pred < 0.0)     otw_pred = 0.0;
    if (otw_pred > 65535.0) otw_pred = 65535.0;
    return (uint16_t)(otw_pred + 0.5);
}

/**
 * @brief Calculate silicon speed_idx from the closed-loop OTW read back from
 *        DPLL_REG_OTW_CUR (0x08) after locking at f_dco_hz.
 */
static inline int32_t dpll_calc_speed_idx(double f_dco_hz, uint16_t otw_locked) {
    int32_t lo = 2000, hi = 25000;
    for (int iter = 0; iter < 20; iter++) {
        int32_t mid = (lo + hi) / 2;
        uint16_t pred = dpll_predict_otw(f_dco_hz, mid);
        if (pred > otw_locked) {
            lo = mid;
        } else {
            hi = mid;
        }
    }
    uint16_t p_lo = dpll_predict_otw(f_dco_hz, lo);
    uint16_t p_hi = dpll_predict_otw(f_dco_hz, hi);
    int32_t diff_lo = (int32_t)p_lo - (int32_t)otw_locked;
```

```
    int32_t diff_hi = (int32_t)otw_locked - (int32_t)p_hi;
    return (diff_lo <= diff_hi) ? lo : hi;
}

/**
 * @brief Generate all DPLL setup parameters from reference frequency (f_ref_hz),
 *        target frequency (f_target_hz), and optional silicon speed_idx (0 = Typical).
 */
static inline bool dpll_generate_setup(
    double         f_ref_hz,
    double         f_target_hz,
    int32_t        speed_idx_in,
    double         duty_pct,
    bool           force_mul2,
    dpll_config_t *cfg,
    double        *out_f_dco_hz
) {
    if (!cfg || f_ref_hz < 1.0e6 || f_ref_hz > 10.0e6 || f_target_hz < 1.0 ||
        f_target_hz > 1000.0e6) {
        return false;
    }
    double f_dco_hz = 500.0e6;
    if (f_target_hz > 500.0e6 || (force_mul2 && f_target_hz >= 250.0e6)) {
        cfg->out_mul2_en        = true;
        cfg->out_div_bypass     = true;
        cfg->out_prescale_shift = 0u;
        cfg->out_div_int        = 1u;
        cfg->out_div_frac       = 0u;
        f_dco_hz                = 0.5 * f_target_hz;
    } else if (f_target_hz >= 250.0e6) {
        cfg->out_mul2_en        = false;
        cfg->out_div_bypass     = true;
        cfg->out_prescale_shift = 0u;
        cfg->out_div_int        = 1u;
        cfg->out_div_frac       = 0u;
        f_dco_hz                = f_target_hz;
    } else {
        cfg->out_mul2_en    = false;
        cfg->out_div_bypass = false;
        if (f_target_hz >= 62.499e6 && f_target_hz <= 62.501e6) {
            f_dco_hz = 250.0e6;
        }
        double total_div = f_dco_hz / f_target_hz;
        uint8_t s = 0u;
        while (s < 20u && (total_div / (double)(1ULL << s)) >= 1024.0) {
            s++;
        }
        double scaled_div = total_div / (double)(1ULL << s);
        uint16_t m_int = (uint16_t)scaled_div;
        if (m_int < 1u) m_int = 1u;
        uint16_t f_frac = (uint16_t)((scaled_div - (double)m_int) * 256.0 + 0.5);
        if (f_frac >= 256u) {
            m_int++;
            f_frac = 0u;
        }
        cfg->out_prescale_shift = s;
        cfg->out_div_int        = m_int;
        cfg->out_div_frac       = (uint8_t)f_frac;
    }

    int duty_w = (int)((duty_pct / 100.0) * 256.0 + 0.5);
    if (duty_w < 1)   duty_w = 1;
    if (duty_w > 255) duty_w = 255;
    cfg->out_duty = (uint8_t)duty_w;
```

```
    double n_total = f_dco_hz / f_ref_hz;
    uint16_t div_n = (uint16_t)(n_total + 1e-9);
    uint16_t fb_f  = (uint16_t)((n_total - (double)div_n) * 256.0 + 0.5);
    if (fb_f >= 256u) {
        div_n++;
        fb_f = 0u;
    }
    cfg->div_n       = div_n;
    cfg->fb_frac     = (uint8_t)fb_f;
    cfg->otw_nominal = dpll_predict_otw(f_dco_hz, speed_idx_in);

    if (cfg->fb_frac != 0u) {
        cfg->kp_mult           = 4u;
        cfg->ki_mult           = 1u;
        cfg->ki_shift          = 3u;
        cfg->fast_lock_en      = false;
        cfg->fast_lock_mult    = 1u;
        cfg->gear_shift_thresh = 48u;
    } else if (f_ref_hz <= 2.0e6) {
        cfg->kp_mult           = 4u;
        cfg->ki_mult           = 1u;
        cfg->ki_shift          = (cfg->otw_nominal < 0x7E00u) ? 4u : 3u;
        cfg->fast_lock_en      = (cfg->otw_nominal < 0x7E00u) ? false : true;
        cfg->fast_lock_mult    = 1u;
        cfg->gear_shift_thresh = 24u;
    } else {
        cfg->kp_mult           = 8u;
        cfg->ki_mult           = 1u;
        cfg->ki_shift          = 1u;
        cfg->fast_lock_en      = true;
        cfg->fast_lock_mult    = 2u;
        cfg->gear_shift_thresh = 48u;
    }
    cfg->integrator_freeze = false;
    cfg->integrator_rst    = false;
    cfg->clr_lock          = false;

    if (out_f_dco_hz) {
        *out_f_dco_hz = f_dco_hz;
    }
    return true;
}

/**
 * @brief Compute calibrated feed-forward nominal OTW for a target DCO frequency
     (250..500 MHz)
 */
static inline uint16_t dpll_calc_otw_nominal(uint16_t dco_freq_mhz) {
    return dpll_predict_otw((double)dco_freq_mhz * 1.0e6, DPLL_SPEED_UNCALIBRATED);
}

/**
 * @brief Serialize a dpll_config_t structure into an 8-byte (64-bit) MSB-first SPI
     core burst buffer
 */
static inline void dpll_pack_burst64(const dpll_config_t *cfg, uint8_t buf[8]) {
    uint16_t dn = cfg->div_n & 0x03FFu;
    buf[0] = DPLL_CMD_BURST_WRITE;                                  /* [63:56] 0xFF
        Header */
    buf[1] = (uint8_t)((cfg->otw_nominal >> 8) & 0xFFu);            /* [55:48]
        otw_nominal[15:8] */
    buf[2] = (uint8_t)(cfg->otw_nominal & 0xFFu);                   /* [47:40]
        otw_nominal[7:0] */
    buf[3] = (uint8_t)((dn >> 2) & 0xFFu);                          /* [39:32] div_n
        [9:2] */
```

```
    buf[4] = (uint8_t)(((dn & 0x03u) << 6)                          /* [31:30] div_n
        [1:0] */
                     | (cfg->kp_mult & 0x3Fu));                     /* [29:24] kp_mult
                        [5:0] */
    buf[5] = (uint8_t)(((cfg->ki_mult & 0x0Fu) << 4)                /* [23:20] ki_mult
        [3:0] */
                     | (cfg->ki_shift & 0x0Fu));                    /* [19:16] ki_shift
                        [3:0] */
    buf[6] = (uint8_t)(((cfg->fast_lock_en ? 1u : 0u) << 7)         /* [15]
        fast_lock_en */
                     | ((cfg->fast_lock_mult & 0x07u) << 4)         /* [14:12]
                        fast_lock_mult[2:0] */
                     | ((cfg->gear_shift_thresh >> 4) & 0x0Fu));    /* [11:8]
                        gear_shift_thresh[7:4] */
    buf[7] = (uint8_t)(((cfg->gear_shift_thresh & 0x0Fu) << 4)      /* [7:4]
        gear_shift_thresh[3:0] */
                     | ((cfg->integrator_freeze ? 1u : 0u) << 3)    /* [3]
                        integrator_freeze */
                     | ((cfg->integrator_rst    ? 1u : 0u) << 2)    /* [2]
                        integrator_rst */
                     | ((cfg->clr_lock          ? 1u : 0u) << 1)    /* [1]     clr_lock
                        */
                     | (cfg->out_mul2_en        ? 1u : 0u));        /* [0]
                        out_mul2_en */
}

/**
 * @brief Serialize a dpll_config_t structure into a 12-byte (96-bit) extended full-
     chip burst buffer
 */
static inline void dpll_pack_burst96(const dpll_config_t *cfg, uint8_t buf[12]) {
    uint16_t m_int = cfg->out_div_int & 0x03FFu;
    dpll_pack_burst64(cfg, buf);
    buf[8]  = (uint8_t)(((cfg->out_div_bypass ? 1u : 0u) << 7)
                      | ((cfg->out_prescale_shift & 0x1Fu) << 2)
                      | ((m_int >> 8) & 0x03u));
    buf[9]  = (uint8_t)(m_int & 0xFFu);
    buf[10] = cfg->out_duty;
    buf[11] = cfg->out_div_frac;
}

/* ========================================================================== */
/* 4. Arduino Hardware SPI Transfer Functions (SPI Mode 0, MSB-First)         */
/* ========================================================================== */

#ifdef ARDUINO

#ifndef DPLL_SPI_CLOCK_HZ
#define DPLL_SPI_CLOCK_HZ (4000000UL) /**< Default 4 MHz SPI clock (supports up to 20
    MHz) */
#endif

/**
 * @brief Write a 16-bit value to a 7-bit DPLL register via a 24-bit SPI Mode 0
     transaction
 */
static inline void dpll_spi_write_reg24(uint8_t cs_pin, uint8_t reg_addr, uint16_t data
    ) {
    SPI.beginTransaction(SPISettings(DPLL_SPI_CLOCK_HZ, MSBFIRST, SPI_MODE0));
    digitalWrite(cs_pin, LOW);
    SPI.transfer((uint8_t)(DPLL_SPI_RW_WRITE_BIT | (reg_addr & DPLL_SPI_ADDR_MASK)));
    SPI.transfer((uint8_t)((data >> 8) & 0xFFu));
    SPI.transfer((uint8_t)(data & 0xFFu));
    digitalWrite(cs_pin, HIGH);
    SPI.endTransaction();
```

```
}

/**
 * @brief Read a 16-bit value from a 7-bit DPLL register via a 24-bit SPI Mode 0
     transaction
 */
static inline uint16_t dpll_spi_read_reg24(uint8_t cs_pin, uint8_t reg_addr) {
    uint8_t hi, lo;
    SPI.beginTransaction(SPISettings(DPLL_SPI_CLOCK_HZ, MSBFIRST, SPI_MODE0));
    digitalWrite(cs_pin, LOW);
    SPI.transfer((uint8_t)(reg_addr & DPLL_SPI_ADDR_MASK)); /* Bit 7 = 0 (Read) */
    hi = SPI.transfer(0x00u);
    lo = SPI.transfer(0x00u);
    digitalWrite(cs_pin, HIGH);
    SPI.endTransaction();
    return ((uint16_t)hi << 8) | (uint16_t)lo;
}

/**
 * @brief Atomically program core DPLL parameters via a 64-bit (8-byte) SPI burst write
 */
static inline void dpll_spi_write_burst64(uint8_t cs_pin, const dpll_config_t *cfg) {
    uint8_t frame[8];
    dpll_pack_burst64(cfg, frame);
    SPI.beginTransaction(SPISettings(DPLL_SPI_CLOCK_HZ, MSBFIRST, SPI_MODE0));
    digitalWrite(cs_pin, LOW);
    for (uint8_t i = 0; i < 8u; i++) {
        SPI.transfer(frame[i]);
    }
    digitalWrite(cs_pin, HIGH);
    SPI.endTransaction();
}

/**
 * @brief Atomically program all core and output divider parameters via a 96-bit (12-
     byte) SPI burst write
 */
static inline void dpll_spi_write_burst96(uint8_t cs_pin, const dpll_config_t *cfg) {
    uint8_t frame[12];
    dpll_pack_burst96(cfg, frame);
    SPI.beginTransaction(SPISettings(DPLL_SPI_CLOCK_HZ, MSBFIRST, SPI_MODE0));
    digitalWrite(cs_pin, LOW);
    for (uint8_t i = 0; i < 12u; i++) {
        SPI.transfer(frame[i]);
    }
    digitalWrite(cs_pin, HIGH);
    SPI.endTransaction();
}

/**
 * @brief Read REG_STATUS (0x04) and verify the 0x15 magic signature, lock status, and
     10-bit div_n
 */
static inline bool dpll_read_status(uint8_t cs_pin, bool *is_locked, uint16_t *
    current_div_n) {
    uint16_t raw = dpll_spi_read_reg24(cs_pin, DPLL_REG_STATUS);
    uint8_t magic = (uint8_t)((raw >> DPLL_STATUS_MAGIC_SHIFT) & 0x1Fu);
    if (is_locked) {
        *is_locked = (raw & DPLL_STATUS_LOCKED_BIT) != 0u;
    }
    if (current_div_n) {
        *current_div_n = (uint16_t)(raw & DPLL_STATUS_DIV_N_MASK);
    }
    return (magic == DPLL_STATUS_MAGIC_EXPECTED);
}
```

```
/**
 * @brief Full-Chip Firmware Configuration, Lock Acquisition & Speed-Index Calibration:
 *        Takes reference frequency f_ref_hz (1..10 MHz), target frequency f_target_hz
 *        (1 Hz .. 1000 MHz), and optional speed_idx_in (0 = Typical corner default).
 *        Programs all setup parameters over SPI, acquires lock, reads back REG_OTW_CUR
 *        (0x08), and returns the calibrated silicon speed_idx so future calls passing
 *        this speed_idx lock maximally fast.
 */
static inline int32_t dpll_configure_and_lock(
    uint8_t        cs_pin,
    uint8_t        lock_pin,
    double         f_ref_hz,
    double         f_target_hz,
    int32_t        speed_idx_in,
    double         duty_pct,
    bool           force_mul2,
    uint16_t      *out_otw_locked,
    uint32_t      *out_lock_us
) {
    dpll_config_t cfg;
    double        f_dco_hz = 500.0e6;
    if (!dpll_generate_setup(f_ref_hz, f_target_hz, speed_idx_in, duty_pct, force_mul2,
         &cfg, &f_dco_hz)) {
        return -1;
    }

    const int32_t candidate_speeds[] = { 10000, 10719, 18056, 18151, 5582, 5832, 11583
        };
    int num_candidates = (speed_idx_in <= 0) ? 7 : 1;

    uint32_t t0 = micros();
    dpll_spi_write_reg24(cs_pin, DPLL_REG_FB_FRAC, dpll_pack_fb_frac(cfg.fb_frac, cfg.
        out_mul2_en));
    dpll_spi_write_reg24(cs_pin, DPLL_REG_OUT_DIV2, dpll_pack_out_div2(cfg.out_duty,
        cfg.out_div_frac));
    dpll_spi_write_reg24(cs_pin, DPLL_REG_OUT_DIV1, dpll_pack_out_div1(cfg.
        out_div_bypass, cfg.out_prescale_shift, cfg.out_div_int));

    for (int attempt = 0; attempt < num_candidates; attempt++) {
        if (attempt > 0) {
            cfg.otw_nominal = dpll_predict_otw(f_dco_hz, candidate_speeds[attempt]);
        }
        dpll_spi_write_burst64(cs_pin, &cfg);
        uint32_t wait_t0 = micros();
        uint32_t timeout_us = (uint32_t)((40.0 * 1.0e6) / f_ref_hz) + 10U;
        while (digitalRead(lock_pin) == LOW && (micros() - wait_t0) < timeout_us) {
            /* wait for lock_pin */
        }
        if (digitalRead(lock_pin) == HIGH) {
            break;
        }
    }

    if (out_lock_us) {
        *out_lock_us = micros() - t0;
    }
    uint16_t otw_cur = dpll_spi_read_reg24(cs_pin, DPLL_REG_OTW_CUR);
    if (out_otw_locked) {
        *out_otw_locked = otw_cur;
    }
    return dpll_calc_speed_idx(f_dco_hz, otw_cur);
}

#endif /* ARDUINO */
```

```
#ifdef __cplusplus
}
#endif

#endif /* DPLL_SPI_H */
```

Listing 4: Portable C/C++ Arduino SPI Driver Header (`dpll_spi.h`).

A.4 Arduino Programming Example (`dpll_arduino_example.ino`)

Listing 5 presents the complete Arduino sketch (`testchip/firmware/dpll_arduino_example/dpll_arduino_example.ino`) demonstrating initial Typical-corner lock, `REG_OTW_CUR` (`0x08`) speed-index calibration, and maximal-fast calibrated locking across 50 MHz, 500 MHz, 1.0 GHz (2× XOR), 375 MHz (Fractional-N), and 250 MHz.

```
/**
 * @file dpll_arduino_example.ino
 * @brief Complete Arduino Programming Example for the IHP 130nm SG13CMOS5L DPLL
    Breakout Board
 *
 * Hardware Connections (3.3V Logic Arduino -> 20x30mm DPLL Breakout Board):
 *   - Arduino 3.3V   ---> Pin J11 (IOV / 3.3V Input: powers IO ring, Blue LED D1, and
    U2/U3 TLV773 1.2V LDOs)
 *   - Arduino GND    ---> Pin J10 / J6 (GND: solid B.Cu ground plane)
 *   - Arduino Pin 10 ---> Pin J3  (CS_N: Active-Low SPI Chip Select)
 *   - Arduino SCK    ---> Pin J4  (SCK : Asynchronous SPI Clock, up to 20 MHz)
 *   - Arduino MOSI   ---> Pin J1  (MOSI: SPI Master-Out Slave-In)
 *   - Arduino MISO   ---> Pin J5  (MISO: SPI Master-In Slave-Out)
 *   - Arduino Pin 7  ---> Pin J2  (RSTN: Active-Low Chip Reset)
 *   - Arduino Pin 6  <--- Pin J7  (LCK : Hardware Lock Status Output, also drives Red
    LED D2)
 *   - 1..10 MHz Ref  ---> Pin J2  (RCLK: 1..10 MHz 3.3V CMOS Reference Clock Input,
    5.0 MHz nominal)
 *   - RF Output      <--- J1 SMA  (50-Ohm Coaxial Output: 1 Hz .. 500 MHz + 500 MHz ..
     1.0 GHz 2x XOR Mode)
 */

#include <SPI.h>
#include "../dpll_spi.h"

/* Pin definitions on the Arduino */
static const uint8_t PIN_DPLL_CS_N   = 10;
static const uint8_t PIN_DPLL_RST_N  = 7;
static const uint8_t PIN_DPLL_LOCKED = 6;

/**
 * @brief Wait for the DPLL hardware lock pin (and Red LED D2) to assert HIGH
 */
static bool wait_for_dpll_lock(uint32_t timeout_us, uint32_t *elapsed_us) {
    uint32_t t0 = micros();
    while (digitalRead(PIN_DPLL_LOCKED) == LOW) {
        if ((micros() - t0) >= timeout_us) {
            if (elapsed_us) *elapsed_us = timeout_us;
            return false;
        }
    }
    if (elapsed_us) *elapsed_us = micros() - t0;
    return true;
}

/**
 * @brief Print all readable DPLL registers over Serial
 */
```

```
static void print_dpll_telemetry(const char *label, int32_t speed_idx, uint32_t lock_us
    ) {
    uint16_t otw_nom   = dpll_spi_read_reg24(PIN_DPLL_CS_N, DPLL_REG_OTW_NOM);
    uint16_t otw_cur   = dpll_spi_read_reg24(PIN_DPLL_CS_N, DPLL_REG_OTW_CUR);
    uint16_t div_ctrl  = dpll_spi_read_reg24(PIN_DPLL_CS_N, DPLL_REG_DIV_CTRL);
    uint16_t out_div1  = dpll_spi_read_reg24(PIN_DPLL_CS_N, DPLL_REG_OUT_DIV1);
    uint16_t out_div2  = dpll_spi_read_reg24(PIN_DPLL_CS_N, DPLL_REG_OUT_DIV2);
    uint16_t fb_frac   = dpll_spi_read_reg24(PIN_DPLL_CS_N, DPLL_REG_FB_FRAC);
    bool     spi_locked = false;
    uint16_t div_n      = 0;
    bool     magic_ok   = dpll_read_status(PIN_DPLL_CS_N, &spi_locked, &div_n);
    bool     pin_locked = (digitalRead(PIN_DPLL_LOCKED) == HIGH);

    Serial.println(F("-----------------------------------------------------------------"));
    Serial.print(F("[DPLL] Stage: "));
    Serial.println(label);
    Serial.print(F("  REG_OTW_NOM (00) : 0x"));
    Serial.print(otw_nom, HEX);
    Serial.print(F(" | REG_OTW_CUR (08): 0x"));
    Serial.print(otw_cur, HEX);
    Serial.print(F(" | speed_idx="));
    Serial.print(speed_idx);
    Serial.print(F(" | t_lock="));
    Serial.print(lock_us);
    Serial.println(F(" us"));
    Serial.print(F("  REG_DIV_CTRL(01) : 0x"));
    Serial.print(div_ctrl, HEX);
    Serial.print(F(" (N="));
    Serial.print(div_n);
    Serial.print(F(", F_fb="));
    Serial.print(fb_frac & 0xFFu);
    Serial.println(F("/256)"));
    Serial.print(F("  REG_OUT_DIV1(05) : 0x"));
    Serial.print(out_div1, HEX);
    Serial.print(F(" | REG_OUT_DIV2(06): 0x"));
    Serial.print(out_div2, HEX);
    Serial.print(F(" | 2x_XOR="));
    Serial.println((div_ctrl & DPLL_DIV_OUT_MUL2_EN_BIT) ? F("ON (500-1000 MHz)") : F("
        OFF"));
    Serial.print(F("  Lock Status      : PIN_LOCKED="));
    Serial.print(pin_locked ? F("1 (Red LED ON)") : F("0 (OFF)"));
    Serial.print(F(" | SPI_LOCKED="));
    Serial.print(spi_locked ? F("1") : F("0"));
    Serial.print(F(" | Magic(0x15)="));
    Serial.println(magic_ok ? F("OK") : F("ERR"));
}

void setup() {
    Serial.begin(115200);
    while (!Serial && millis() < 3000UL) {
        /* Wait up to 3s for USB Serial Monitor */
    }

    pinMode(PIN_DPLL_CS_N, OUTPUT);
    digitalWrite(PIN_DPLL_CS_N, HIGH);
    pinMode(PIN_DPLL_RST_N, OUTPUT);
    digitalWrite(PIN_DPLL_RST_N, LOW);
    pinMode(PIN_DPLL_LOCKED, INPUT);

    SPI.begin();

    Serial.println(F("\n================================================================"))
        ;
    Serial.println(F("  IHP 130nm SG13CMOS5L All-Digital PLL Breakout Controller"));
    Serial.println(F("================================================================"));
```

```
    /* 1. Release active-low reset (RST_N) after power rails settle */
    delay(2);
    digitalWrite(PIN_DPLL_RST_N, HIGH);

    uint16_t otw_locked = 0;
    uint32_t lock_us    = 0;

    /* 2. First Call: Pass speed_idx = DPLL_SPEED_UNCALIBRATED (0) to start from
     *    the real chip Typical corner (0xEE89), lock to 50.0 MHz from a 5.0 MHz
     *    reference clock, read back REG_OTW_CUR (0x08), and obtain chip_speed_idx! */
    int32_t chip_speed_idx = dpll_configure_and_lock(
        PIN_DPLL_CS_N, PIN_DPLL_LOCKED,
        5.0e6, 50.0e6, DPLL_SPEED_UNCALIBRATED, 50.0, false,
        &otw_locked, &lock_us
    );
    print_dpll_telemetry("Initial Lock & Speed Calibration: 50.0 MHz Output (Ref=5 MHz)
        ", chip_speed_idx, lock_us);

    /* 3. Subsequent Call #1: Pass calibrated chip_speed_idx for maximal fast lock
     *    at 500.0 MHz direct DCO output */
    chip_speed_idx = dpll_configure_and_lock(
        PIN_DPLL_CS_N, PIN_DPLL_LOCKED,
        5.0e6, 500.0e6, chip_speed_idx, 50.0, false,
        &otw_locked, &lock_us
    );
    print_dpll_telemetry("Fast Calibrated Lock: 500.0 MHz Direct DCO Output",
        chip_speed_idx, lock_us);

    /* 4. Subsequent Call #2: Pass calibrated chip_speed_idx for 1.000 GHz (1000 MHz)
     *    2x XOR output multiplier mode */
    chip_speed_idx = dpll_configure_and_lock(
        PIN_DPLL_CS_N, PIN_DPLL_LOCKED,
        5.0e6, 1000.0e6, chip_speed_idx, 50.0, true,
        &otw_locked, &lock_us
    );
    print_dpll_telemetry("Fast Calibrated Lock: 1.000 GHz (2x XOR Multiplier)",
        chip_speed_idx, lock_us);

    /* 5. Subsequent Call #3: Pass calibrated chip_speed_idx for 375.0 MHz Fractional-N
     *    DCO synthesis (N = 75.0 at 5 MHz ref, or N = 37.5 at 10 MHz ref) */
    chip_speed_idx = dpll_configure_and_lock(
        PIN_DPLL_CS_N, PIN_DPLL_LOCKED,
        5.0e6, 375.0e6, chip_speed_idx, 50.0, false,
        &otw_locked, &lock_us
    );
    print_dpll_telemetry("Fast Calibrated Lock: 375.0 MHz DCO Output", chip_speed_idx,
        lock_us);
}

void loop() {
    delay(1000);
}
```

Listing 5: Arduino Sketch (`dpll_arduino_example.ino`) for Controlling the Standalone DPLL Testchip.